

Basic Electrical Engineering

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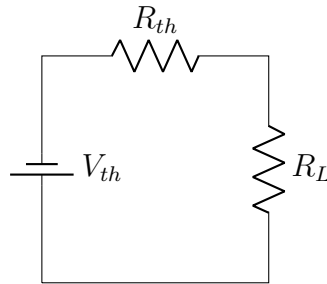
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Network Theorems with Circuit Diagrams

1. Thevenin's Theorem

Statement: Any linear, bilateral network containing voltage sources and resistances can be replaced by an equivalent circuit consisting of a single voltage source V_{th} in series with a resistance R_{th} , connected to the load.

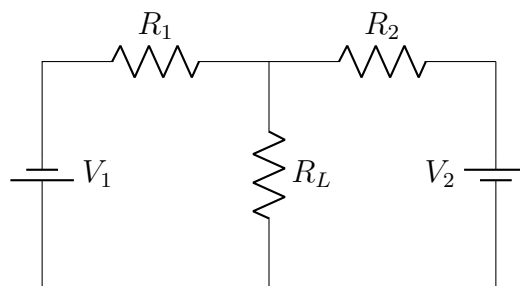


Explanation:

- V_{th} : Open-circuit voltage across the load terminals.
- R_{th} : Equivalent resistance seen from the load terminals with sources replaced by internal resistances.

2. Superposition Theorem

Statement: In a linear network with multiple independent sources, the response in any element is the algebraic sum of the responses produced by each source acting alone, with all others replaced by their internal resistances.

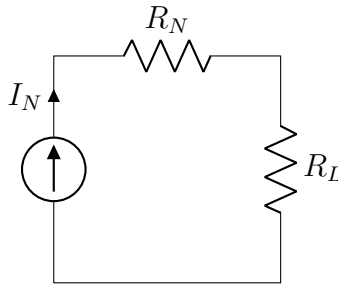


Explanation:

- Consider one source at a time: short-circuit all voltage sources and open-circuit all current sources.
- Add all individual responses to get total current or voltage.

3. Norton's Theorem

Statement: Any linear network can be replaced by a single current source I_N in parallel with a resistance R_N , connected to the load.



Explanation:

- I_N : Short-circuit current through load terminals.
- $R_N = R_{th}$: Equivalent resistance of the network.

4. Maximum Power Transfer Theorem

Statement: Maximum power is transferred from a source to a load when the load resistance equals the source's internal resistance.

$$R_L = R_s$$

Maximum Power:

$$P_{\max} = \frac{V^2}{4R_s}$$

Note : The word “bilateral” appears in Thevenin’s and Norton’s theorems because those theorems involve replacing the entire network by a single equivalent source and resistance — which assumes that resistance behaves the same in both directions. In Superposition theorem, only linearity is required — so “bilateral” is optional and usually omitted.

Proof of Maximum Power Transfer Theorem

Theorem: Maximum power is transferred from a source to a load when the load resistance R_L equals the source resistance R_s .

Proof

From the circuit, the current through the load is:

$$I = \frac{V}{R_s + R_L}$$

The power delivered to the load R_L is:

$$P_L = I^2 R_L = \left(\frac{V}{R_s + R_L} \right)^2 R_L$$

$$P_L = \frac{V^2 R_L}{(R_s + R_L)^2}$$

To find the condition for maximum power, differentiate P_L with respect to R_L and set to zero:

$$\frac{dP_L}{dR_L} = \frac{V^2 [(R_s + R_L)^2 - 2R_L(R_s + R_L)]}{(R_s + R_L)^4} = 0$$

Simplify the bracket:

$$(R_s + R_L)^2 - 2R_L(R_s + R_L) = (R_s + R_L) [(R_s + R_L) - 2R_L] = (R_s + R_L)(R_s - R_L)$$

Thus:

$$\frac{dP_L}{dR_L} = \frac{V^2(R_s + R_L)(R_s - R_L)}{(R_s + R_L)^4} = 0$$

Since $V \neq 0$ and $R_s + R_L > 0$, we have:

$$R_s - R_L = 0$$

$$\boxed{R_L = R_s}$$

Maximum Power

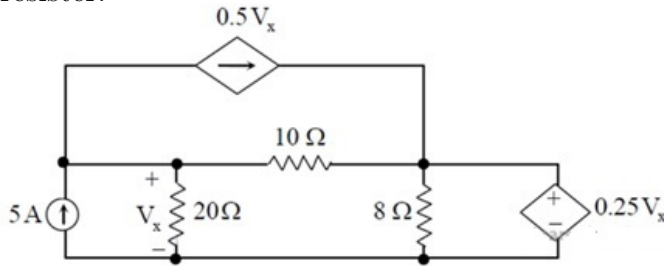
Substitute $R_L = R_s$ into P_L :

$$P_{\max} = \frac{V^2 R_s}{(R_s + R_s)^2} = \frac{V^2 R_s}{(2R_s)^2} = \frac{V^2}{4R_s}$$

$$\boxed{P_{\max} = \frac{V^2}{4R_s}}$$

Maximum power is delivered to the load when $R_L = R_s$, and the maximum value of power is $\frac{V^2}{4R_s}$.

Numerical : Find the current through $20\ \Omega$ resistor and power consumed in $10\ \Omega$ resistor.



Given: The circuit contains a current source of $5A$, a 20Ω resistor across which

$$V_x = V_A$$

and two dependent sources: a current source $0.5V_x$ and a voltage source of value $0.25V_x$. The 10Ω and 8Ω resistors connect between node A and the right-side node.

$$V_B = 0.25V_x = 0.25V_A$$

(1) Current through the 20Ω resistor

Apply KCL at node A :

$$5 = \frac{V_A}{20} + \frac{V_A - V_B}{10} + 0.5V_A$$

Substituting $V_B = 0.25V_A$:

$$5 = \frac{V_A}{20} + \frac{V_A - 0.25V_A}{10} + 0.5V_A = 0.05V_A + 0.075V_A + 0.5V_A$$

$$5 = 0.625V_A \Rightarrow V_A = \frac{5}{0.625} = 8V$$

Thus voltage across the 20Ω resistor is $V_x = V_A = 8V$.

$$I_{20} = \frac{V_x}{20} = \frac{8}{20} = 0.4A$$

So, the current through the 20Ω resistor is:

$$\boxed{I_{20} = 0.4A}$$

(2) Power consumed in the 10Ω resistor

Voltage across the 10Ω resistor:

$$V_{10} = V_A - V_B = V_A - 0.25V_A = 0.75V_A = 0.75 \times 8 = 6V$$

Current through 10Ω :

$$I_{10} = \frac{V_{10}}{10} = \frac{6}{10} = 0.6A$$

Power consumed:

$$P_{10} = I_{10}^2 \cdot 10\Omega = (0.6)^2 \times 10 = 3.6W$$

$$\boxed{P_{10} = 3.6W}$$

$$\boxed{I_{20} = 0.4A, \quad P_{10} = 3.6W}$$

Numerical : A dc voltage source of 120 Volts was connected to R_L series circuit with $R=20\ \Omega$ and $L=8\ \text{H}$. Find (a) the current at $t=0.6\ \text{sec}$. (b) Time at which voltage drops across R and L are same.

Solution : Given: A DC source $V = 120\ \text{V}$ is applied at $t = 0$ to a series RL circuit with

$$R = 20\ \Omega, \quad L = 8\ \text{H}.$$

The time constant for an RL circuit is

$$\tau = \frac{L}{R} = \frac{8}{20} = 0.4\ \text{s}.$$

The steady-state (final) current is

$$I_{\infty} = \frac{V}{R} = \frac{120}{20} = 6\ \text{A}.$$

The transient response of current for a step DC excitation is

$$\boxed{i(t) = I_{\infty} (1 - e^{-t/\tau}) = 6 (1 - e^{-t/0.4})\ \text{A} .}$$

(a) Current at $t = 0.6\ \text{s}$:

$$i(0.6) = 6 (1 - e^{-0.6/0.4}) = 6 (1 - e^{-1.5}) .$$

Using $e^{-1.5} \approx 0.22313016$,

$$i(0.6) \approx 6(1 - 0.22313016) = 6(0.77686984) \approx 4.6612\ \text{A}.$$

$$\boxed{i(0.6\ \text{s}) \approx 4.6612\ \text{A}}$$

(b) Time when voltage across R and L are equal:

Voltage across resistor:

$$v_R(t) = i(t)R = RI_{\infty} (1 - e^{-t/\tau}) .$$

Voltage across inductor:

$$v_L(t) = L \frac{di}{dt} = L \cdot I_{\infty} \cdot \frac{1}{\tau} e^{-t/\tau} .$$

Since $\tau = L/R$, we have $\frac{L}{\tau} = R$. Hence

$$v_L(t) = RI_{\infty}e^{-t/\tau}.$$

Set $v_R(t) = v_L(t)$:

$$RI_{\infty}(1 - e^{-t/\tau}) = RI_{\infty}e^{-t/\tau}.$$

Cancel RI_{∞} (nonzero):

$$1 - e^{-t/\tau} = e^{-t/\tau} \Rightarrow 1 = 2e^{-t/\tau}.$$

Thus

$$e^{-t/\tau} = \frac{1}{2} \Rightarrow -\frac{t}{\tau} = \ln \frac{1}{2} = -\ln 2,$$

so

$$\boxed{t = \tau \ln 2 = 0.4 \ln 2 \approx 0.2773 \text{ s.}}$$

At this time $e^{-t/\tau} = 1/2$, so each voltage equals

$$v_R = v_L = RI_{\infty}(1 - \frac{1}{2}) = \frac{1}{2}RI_{\infty} = \frac{1}{2} \times 20 \times 6 = 60 \text{ V},$$

and $v_R + v_L = 60 + 60 = 120 \text{ V}$ as expected.

Final answers:

$$\boxed{i(0.6 \text{ s}) \approx 4.6612 \text{ A}, \quad t_{\text{equal}} = 0.4 \ln 2 \approx 0.2773 \text{ s.}}$$

Numerical : A resistance of 30 ohm, inductance of 0.2 H and capacitance of 150 μF connected in series and fed by a 230V, 50 Hz Supply. Find $X_L, X_C, Z, Y, \text{p.f., active power}$ and reactive power.

Solution : Given:

$$R = 30 \Omega, \quad L = 0.2 \text{ H}, \quad C = 150 \mu\text{F} = 150 \times 10^{-6} \text{ F}, \quad f = 50 \text{ Hz}, \quad V_{\text{rms}} = 230 \text{ V}.$$

Angular frequency:

$$\omega = 2\pi f = 2\pi(50) = 100\pi \text{ rad/s}.$$

Reactances:

$$X_L = \omega L = 2\pi f L = 100\pi \times 0.2 = 20\pi \approx 62.8319 \Omega,$$

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C} = \frac{1}{100\pi \times 150 \times 10^{-6}} \approx 21.2207 \Omega.$$

$$\boxed{X_L \approx 62.832 \Omega, \quad X_C \approx 21.221 \Omega}$$

Net reactance (series):

$$X = X_L - X_C \approx 62.8319 - 21.2207 = 41.6112 \Omega.$$

Impedance (complex and magnitude):

$$Z = R + jX = 30 + j 41.6112 \, \Omega,$$

$$|Z| = \sqrt{R^2 + X^2} = \sqrt{30^2 + 41.6112^2} \approx \sqrt{900 + 1731.79} \approx 51.295 \, \Omega.$$

$$\boxed{|Z| \approx 51.295 \, \Omega}$$

Admittance magnitude:

$$|Y| = \frac{1}{|Z|} \approx \frac{1}{51.295} \approx 0.01949 \, \text{S}.$$

$$\boxed{|Y| \approx 0.01949 \, \text{S}}$$

Circuit current (rms):

$$I_{\text{rms}} = \frac{V_{\text{rms}}}{|Z|} = \frac{230}{51.295} \approx 4.485 \, \text{A}.$$

$$\boxed{I_{\text{rms}} \approx 4.485 \, \text{A}}$$

Power factor and phase angle:

$$\cos \varphi = \frac{R}{|Z|} = \frac{30}{51.295} \approx 0.5849 \quad (\text{lagging, inductive}),$$

$$\varphi = \cos^{-1}(0.5849) \approx 54.18^\circ.$$

$$\boxed{\text{pf} \approx 0.5849 \, (\text{lagging})}$$

Active (real) power:

$$P = I_{\text{rms}}^2 R \approx (4.485)^2 \times 30 \approx 20.113 \times 30 \approx 603.4 \, \text{W}.$$

$$\boxed{P \approx 603.4 \, \text{W}}$$

Reactive power (net, inductive):

$$Q = I_{\text{rms}}^2 X \approx (4.485)^2 \times 41.6112 \approx 20.113 \times 41.6112 \approx 837.7 \, \text{VAR}.$$

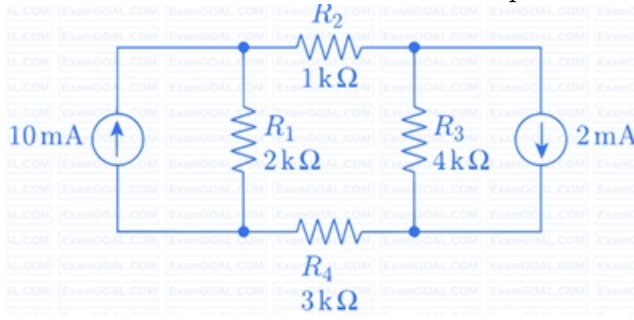
$$\boxed{Q \approx +837.7 \, \text{VAR} \quad (\text{positive} = \text{inductive})}$$

Numerical summary:

$$X_L \approx 62.832 \, \Omega, \quad X_C \approx 21.221 \, \Omega, \quad |Z| \approx 51.295 \, \Omega, \quad |Y| \approx 0.01949 \, \text{S},$$

$$\text{pf} \approx 0.5849 \, (\text{lagging}), \quad P \approx 603.4 \, \text{W}, \quad Q \approx 837.7 \, \text{VAR}.$$

Numerical : Find the Thevenin's equivalent across R_2 .



Solution : Problem: Find the Thevenin equivalent seen across resistor R_2 (the $1\text{ k}\Omega$ resistor) in the circuit shown.

Solution:

Remove R_2 and denote the open-circuit top nodes as A (left top) and B (right top). Let the corresponding bottom nodes be C (under R_1) and D (under R_3). Given: $I_{s1} = 10\text{ mA}$ (left current source, arrow upward from C to A), $I_{s2} = 2\text{ mA}$ (right current source, arrow downward from B to D), $R_1 = 2\text{ k}\Omega$, $R_3 = 4\text{ k}\Omega$, $R_4 = 3\text{ k}\Omega$.

(1) Open-circuit voltage $V_{th} = V_A - V_B$.

Because R_2 is removed, node A is connected only to C through R_1 . KCL at node A simply forces the current through R_1 to equal the left source current:

$$I_{R_1} = I_{s1} = 10\text{ mA}.$$

Hence the voltage drop across R_1 is

$$V_A - V_C = I_{R_1} R_1 = (10 \times 10^{-3})(2 \times 10^3) = 20\text{ V}.$$

Thus

$$V_A = V_C + 20. \quad (1)$$

Similarly, at the right branch the right current source enforces 2 mA downward from B to D . Because B is connected only to D through R_3 , the resistor current must be

$$I_{R_3} = -I_{s2} = -2\text{ mA}$$

(which means 2 mA flows from D to B). Hence

$$V_B - V_D = I_{R_3} R_3 = (-2 \times 10^{-3})(4 \times 10^3) = -8\text{ V},$$

or

$$V_B = V_D - 8. \quad (2)$$

Now write KCL at bottom node C . Currents leaving C are: $-I_{s1} = 10\text{ mA}$ (upwards through the left current source), $-I_{C \rightarrow D} = \frac{V_C - V_D}{R_4}$ (through R_4), $-I_{C \rightarrow A} = \frac{V_C - V_A}{R_1}$ (through R_1).

KCL: $I_{s1} + \frac{V_C - V_D}{R_4} + \frac{V_C - V_A}{R_1} = 0$. Substitute $V_A = V_C + 20$ into $\frac{V_C - V_A}{R_1} = \frac{V_C - (V_C + 20)}{R_1} = -\frac{20}{2000} = -10\text{ mA}$. Thus the KCL at C becomes

$$10\text{ mA} + \frac{V_C - V_D}{R_4} - 10\text{ mA} = 0 \implies \frac{V_C - V_D}{R_4} = 0.$$

Hence

$$\boxed{V_C = V_D.} \quad (3)$$

Using (1),(2) and (3) we obtain

$$V_{th} = V_A - V_B = (V_C + 20) - (V_D - 8) = (V_C - V_D) + 28 = 28 \text{ V}.$$

$$\boxed{V_{th} = 28 \text{ V}}$$

(2) Thevenin resistance R_{th} .

Turn off all independent sources: open the two independent current sources. The passive resistor network seen between A and B then becomes a single series path $A-R_1-R_4-R_3-B$. Therefore

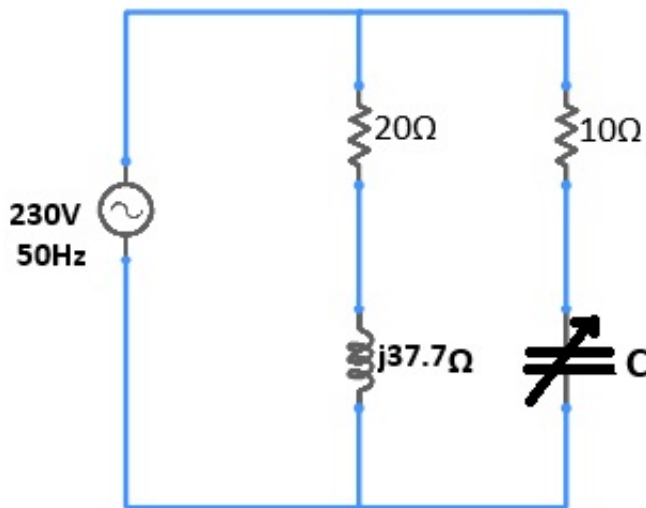
$$R_{th} = R_1 + R_4 + R_3 = 2 \text{ k}\Omega + 3 \text{ k}\Omega + 4 \text{ k}\Omega = 9 \text{ k}\Omega.$$

$$\boxed{R_{th} = 9 \text{ k}\Omega}$$

Thevenin equivalent across R_2 :

$$\boxed{V_{th} = 28 \text{ V}, \quad R_{th} = 9 \text{ k}\Omega.}$$

Numerical : Show that the circuit shown in the figure can have more than one resonant condition. Assume C to be variable.



Solution : Resonance (in the sense of the *parallel* network) occurs when the total imaginary part of the admittance Y_{tot} is zero:

$$\Im\{Y_{tot}\} = 0, \quad Y_{tot} = Y_1 + Y_2.$$

Step 1: write branch admittances

For a series branch $R + jX$ the admittance is

$$\frac{1}{R + jX} = \frac{R - jX}{R^2 + X^2}.$$

Thus for the left (series $20 + j37.7$) branch

$$Y_1 = \frac{1}{20 + j37.7} = \frac{20 - j37.7}{20^2 + 37.7^2},$$

and for the right (series $10 - jX_C$) branch (note $X_C > 0$ and branch reactance is $-X_C$)

$$Y_2 = \frac{1}{10 - jX_C} = \frac{10 + jX_C}{10^2 + X_C^2}.$$

Step 2: imaginary part of total admittance

Take the imaginary parts of Y_1 and Y_2 :

$$\Im\{Y_1\} = -\frac{37.7}{20^2 + 37.7^2}, \quad \Im\{Y_2\} = \frac{X_C}{10^2 + X_C^2}.$$

So the resonance condition $\Im\{Y_1\} + \Im\{Y_2\} = 0$ becomes

$$-\frac{37.7}{20^2 + 37.7^2} + \frac{X_C}{100 + X_C^2} = 0.$$

Rearrange:

$$\frac{X_C}{100 + X_C^2} = \frac{37.7}{20^2 + 37.7^2}.$$

Step 3: convert to a quadratic equation

Compute the right-hand constant:

$$20^2 + 37.7^2 = 400 + 37.7^2 \approx 400 + 1421.29 = 1821.29,$$

$$k \equiv \frac{37.7}{1821.29} \approx 0.02072.$$

Thus

$$\frac{X_C}{100 + X_C^2} = k \implies X_C = k(100 + X_C^2).$$

Bring all terms to one side:

$$kX_C^2 - X_C + 100k = 0.$$

Substitute $k \approx 0.02072$:

$$0.02072 X_C^2 - X_C + 2.072 = 0.$$

This is a quadratic in X_C . Its discriminant is

$$\Delta = (-1)^2 - 4(0.02072)(2.072) = 1 - 4 \cdot 0.02072 \cdot 2.072 \approx 0.82818 > 0,$$

so there are two real roots.

Step 4: compute the two roots

$$X_C = \frac{1 \pm \sqrt{\Delta}}{2 \cdot 0.02072}.$$

Numerically,

$$\sqrt{\Delta} \approx 0.9090,$$

so

$$X_{C1} \approx \frac{1 + 0.9090}{0.04144} \approx \frac{1.9090}{0.04144} \approx 46.07 \, \Omega,$$

$$X_{C2} \approx \frac{1 - 0.9090}{0.04144} \approx \frac{0.0910}{0.04144} \approx 2.197 \, \Omega.$$

Both solutions are positive and therefore physically admissible (they correspond to two different capacitor reactance values at the fixed supply frequency).

Step 5: convert reactances to capacitances at 50 Hz

Using $X_C = \frac{1}{\omega C}$ with $\omega = 2\pi f = 2\pi \cdot 50 \approx 314.159$ rad/s:

$$C_1 = \frac{1}{\omega X_{C1}} \approx \frac{1}{314.159 \times 46.07} \approx 6.91 \times 10^{-5} \text{ F} = 69.1 \, \mu\text{F},$$

$$C_2 = \frac{1}{\omega X_{C2}} \approx \frac{1}{314.159 \times 2.197} \approx 1.449 \times 10^{-3} \text{ F} = 1449 \, \mu\text{F}.$$

Numerical : Find the r.m.s. value of the voltage defined by

$$e(t) = 5 + 5 \sin(314t + \pi/6).$$

Answer:

The given voltage consists of a dc component and an ac sinusoidal component:

$$e(t) = E_{\text{dc}} + E_m \sin(\omega t + \phi),$$

where

$$E_{\text{dc}} = 5 \quad \text{and} \quad E_m = 5.$$

The rms value of the ac component is

$$E_{\text{ac,rms}} = \frac{E_m}{\sqrt{2}} = \frac{5}{\sqrt{2}} = 3.535 \text{ V}.$$

The total rms voltage is therefore

$$E_{\text{rms}} = \sqrt{E_{\text{dc}}^2 + E_{\text{ac,rms}}^2}.$$

Substituting the values,

$$E_{\text{rms}} = \sqrt{5^2 + (3.535)^2} = \sqrt{25 + 12.5} = \sqrt{37.5} = 6.12 \text{ V}.$$

$$\boxed{E_{\text{rms}} = 6.12 \text{ V}}$$

Numerical: In a certain series resonant circuit, the voltages across the components are:

$$V_C = 150 \text{ V}, \quad V_L = 150 \text{ V}, \quad V_R = 50 \text{ V}.$$

Determine the value of the source voltage.

Answer:

In a series resonant circuit, the inductive and capacitive voltages are equal in magnitude and 180° out of phase. Hence,

$$V_L - V_C = 150 - 150 = 0.$$

Therefore, the source voltage is equal to the resistive voltage:

$$V_s = V_R = 50 \text{ V}.$$

$V_s = 50 \text{ V}$

Q. An alternating voltage

$$v(t) = 100 \sin(314t)$$

is applied to a half-wave rectifier in series with a resistance of

$$R = 25 \Omega.$$

The rectifier has infinite resistance in one direction and zero resistance in the other. Determine the rms value of the current drawn from the supply source.

Answer:

The peak value of the input voltage is

$$V_m = 100 \text{ V}.$$

Since the rectifier offers zero resistance in the conducting direction, the peak current is

$$I_m = \frac{V_m}{R} = \frac{100}{25} = 4 \text{ A}.$$

For a half-wave rectified sine wave, the rms current is given by

$$I_{\text{rms}} = \frac{I_m}{2}.$$

Thus,

$$I_{\text{rms}} = \frac{4}{2} = 2 \text{ A}.$$

$I_{\text{rms}} = 2 \text{ A}$

Numerical : The waveform of a non-sinusoidal voltage has a form factor of 1.15 and a peak factor of 1.4. If the peak value of the voltage is 322 V, determine the average value of the voltage.

Answer:

Given:

$$\text{Form factor} = 1.15 = \frac{V_{\text{rms}}}{V_{\text{avg}}}, \quad \text{Peak factor} = 1.4 = \frac{V_{\text{peak}}}{V_{\text{rms}}}, \quad V_{\text{peak}} = 322 \text{ V}.$$

First, compute the rms value:

$$V_{\text{rms}} = \frac{V_{\text{peak}}}{\text{Peak factor}} = \frac{322}{1.4} = 230 \text{ V}.$$

Now compute the average value using the form factor:

$$V_{\text{avg}} = \frac{V_{\text{rms}}}{1.15} = \frac{230}{1.15} = 200 \text{ V}.$$

$$\boxed{V_{\text{avg}} = 200 \text{ V}}$$

Q. The current in a parallel resonant circuit is 10 A. If the inductance L is doubled and the resistance R and capacitance C are halved, determine the new resonant current.

Answer:

The resonant frequency of a parallel resonant circuit is

$$\omega_0 = \frac{1}{\sqrt{LC}}.$$

After changing the component values,

$$L \rightarrow 2L, \quad C \rightarrow \frac{C}{2},$$

so the product becomes

$$LC \rightarrow (2L) \left(\frac{C}{2} \right) = LC.$$

Thus, the resonant frequency remains unchanged.

At resonance, the current drawn from the supply is

$$I_{\text{res}} = \frac{V}{R}.$$

Since the resistance is halved,

$$R \rightarrow \frac{R}{2},$$

the current doubles:

$$I_{\text{new}} = 2 \times 10 = 20 \text{ A}.$$

$$\boxed{I_{\text{new}} = 20 \text{ A}}$$

Numerical : An alternating current of frequency 50 Hz has a maximum value of 100 A. What will be its value after $\frac{1}{600}$ second from the instant the current is zero?

Answer:

For a sinusoidal current starting from zero,

$$i(t) = I_{\max} \sin(\omega t),$$

where

$$I_{\max} = 100 \text{ A}, \quad \omega = 2\pi f = 2\pi(50) = 100\pi \text{ rad/s}.$$

At time

$$t = \frac{1}{600} \text{ s},$$

the instantaneous current is

$$i = 100 \sin(\omega t) = 100 \sin\left(100\pi \times \frac{1}{600}\right) = 100 \sin\left(\frac{\pi}{6}\right).$$

Since

$$\sin\left(\frac{\pi}{6}\right) = \frac{1}{2},$$

we obtain

$$i = 100 \times \frac{1}{2} = 50 \text{ A}.$$

$$\boxed{i = 50 \text{ A}}$$

Numerical: Series RLC Circuit

Given:

$$R = 3 \Omega, \quad X_L = j6 \Omega, \quad X_C = -j2 \Omega, \quad I = 10\angle(-142^\circ) \text{ A}$$

To find: Voltage applied across the combination (V).

Solution

The total impedance of the series circuit is:

$$Z = R + X_L + X_C$$

$$Z = 3 + j6 - j2 = 3 + j4$$

Magnitude and phase angle of Z :

$$|Z| = \sqrt{3^2 + 4^2} = 5 \Omega$$

$$\angle Z = \tan^{-1}\left(\frac{4}{3}\right) = 53.13^\circ$$

Voltage across the combination:

$$V = I \times Z$$

In polar form:

$$|V| = |I| \times |Z| = 10 \times 5 = 50 \text{ V}$$
$$\angle V = -142^\circ + 53.13^\circ = -88.87^\circ$$

$$\boxed{V = 50 \angle (-88.87^\circ) \text{ V}}$$

In rectangular form:

$$V = 50(\cos(-88.87^\circ) + j \sin(-88.87^\circ))$$
$$V = 0.986 - j49.99 \text{ V}$$

$$\boxed{V = 0.99 - j50.0 \text{ V}}$$

Given:

$$Z = (2 + j1 - j0.5) \Omega = (2 + j0.5) \Omega$$
$$V_1 = 10 \angle 0^\circ \text{ V}, \quad V_2 = 5 \angle 45^\circ \text{ V (opposing polarity)}$$

To find: Magnitude and phase angle of current I .

Solution

Since the two voltages are connected in opposition,

$$V_{\text{net}} = V_1 - V_2$$

Convert V_2 to rectangular form:

$$V_2 = 5(\cos 45^\circ + j \sin 45^\circ) = 3.5355 + j3.5355$$

Hence,

$$V_{\text{net}} = 10 - (3.5355 + j3.5355) = 6.4645 - j3.5355 \text{ V}$$

Now, the current through the impedance is:

$$I = \frac{V_{\text{net}}}{Z} = \frac{6.4645 - j3.5355}{2 + j0.5}$$

Multiply numerator and denominator by the complex conjugate of the denominator:

$$I = \frac{(6.4645 - j3.5355)(2 - j0.5)}{(2 + j0.5)(2 - j0.5)}$$

Simplifying,

$$I = \frac{(12.929 - 1.7678) - j(7.071 + 3.2323)}{4.25}$$
$$I = \frac{11.1612 - j10.3033}{4.25} = 2.626 - j2.424 \text{ A}$$

Therefore,

$$|I| = \sqrt{(2.626)^2 + (-2.424)^2} = 3.574 \text{ A}$$
$$\angle I = \tan^{-1} \left(\frac{-2.424}{2.626} \right) = -42.71^\circ$$

$$\boxed{I = 3.574 \angle (-42.71^\circ) \text{ A}}$$

Or in rectangular form:

$$I = 2.626 - j2.424 \text{ A}$$

Variation of Current and Impedance in a Series Resonant Circuit

A series RLC circuit consists of a resistor R , inductor L , and capacitor C connected in series with an AC supply.

Impedance of the Circuit

$$Z = R + j \left(\omega L - \frac{1}{\omega C} \right)$$
$$|Z| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2}$$

Current in the Circuit

$$I = \frac{V}{|Z|} = \frac{V}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2}}$$

At Resonance

Resonance occurs when:

$$X_L = X_C \quad \Rightarrow \quad \omega L = \frac{1}{\omega C}$$

Hence,

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

At resonance:

$$Z = R \quad (\text{minimum}), \quad I = \frac{V}{R} \quad (\text{maximum})$$

Frequency Response

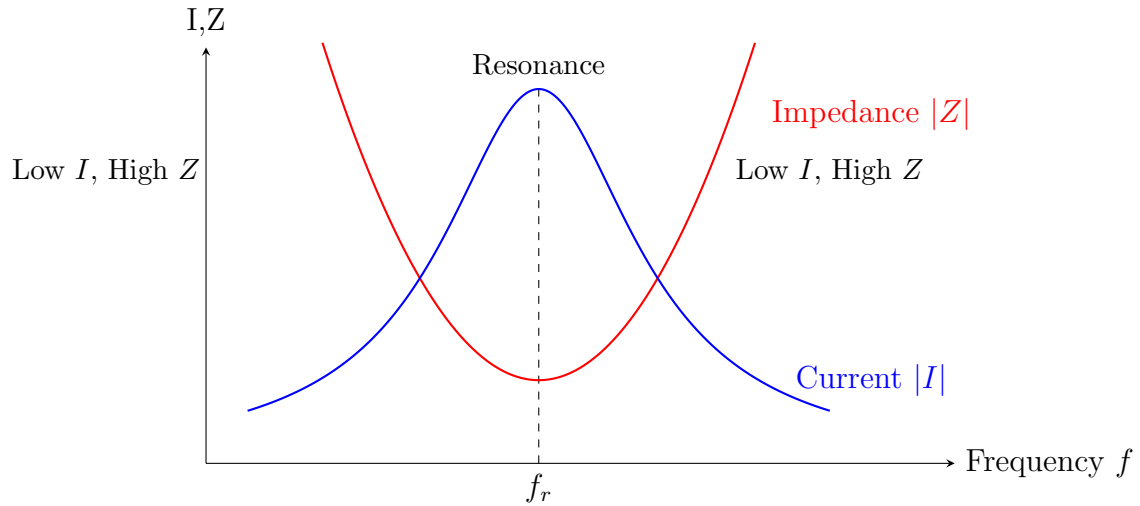
Condition	Frequency	Reactance	Impedance	Current
Below resonance	$f < f_r$	$X_L < X_C$ (capacitive)	High	Low
At resonance	$f = f_r$	$X_L = X_C$ (resistive)	Minimum	Maximum
Above resonance	$f > f_r$	$X_L > X_C$ (inductive)	High	Low

Quality Factor

The sharpness of the resonance curve is defined by the quality factor:

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}}$$

A higher Q indicates a sharper resonance.



Quality Factor (Q-Factor)

Definition

The **Quality Factor (Q)** of a resonant circuit is a dimensionless parameter that measures how sharp or selective the resonance is. It is defined as the ratio of the energy stored to the energy dissipated per cycle of oscillation. One complete oscillation corresponds to 2π radians in angular measure and if multiply by 2π then Energy dissipated per cycle.

$$Q = 2\pi \times \frac{\text{Maximum Energy Stored}}{\text{Energy Dissipated per Cycle}}$$

A higher value of Q indicates a sharper resonance and lower energy loss.

For a Series RLC Circuit

At resonance, the quality factor is given by:

$$Q = \frac{\omega_r L}{R} = \frac{1}{\omega_r C R}$$

where

$$\omega_r = \frac{1}{\sqrt{LC}}$$

is the resonant angular frequency.

For a Parallel RLC Circuit

$$Q = R\sqrt{\frac{C}{L}}$$

In Terms of Bandwidth

The Q-factor can also be expressed as the ratio of resonant frequency to the bandwidth (BW):

$$Q = \frac{f_r}{BW} = \frac{f_r}{f_2 - f_1}$$

where:

- f_r = Resonant frequency,
- f_1, f_2 = Lower and upper half-power frequencies.

A high Q means a narrow bandwidth and a sharp resonance, whereas a low Q indicates a broader resonance curve.

Physical Meaning

- High Q : Energy oscillates many times before decaying (low damping).
- Low Q : Energy decays quickly (high damping).

Applications of Q-Factor

1. **Tuned Circuits:** Used in radio and TV receivers for frequency selection.
2. **Filters:** High-Q circuits are used to design narrow band-pass and notch filters.
3. **Oscillators:** Determines the frequency stability of LC oscillators.
4. **Amplifiers:** In tuned amplifiers, Q affects selectivity and bandwidth.
5. **Measurement:** Helps in accurate determination of L , C , or R using resonance methods.

Example Calculation

Given:

$$L = 10 \text{ mH}, \quad C = 100 \text{ nF}, \quad R = 10 \Omega$$

Then,

$$\omega_r = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(10 \times 10^{-3})(100 \times 10^{-9})}} = 10^5 \text{ rad/s}$$

and

$$Q = \frac{\omega_r L}{R} = \frac{10^5 \times 10^{-2}}{10} = 100$$

Thus, the circuit has a high quality factor, indicating a sharp and selective resonance.

Quality Factor (Q) — Voltage Ratio Definition

In a *series* RLC circuit at resonance, the quality factor Q can be expressed as the ratio of the voltage across the inductor (or capacitor) to the applied (supply) voltage:

$$Q = \frac{V_L}{V_{\text{applied}}} = \frac{V_C}{V_{\text{applied}}}.$$

For a series RLC circuit at resonance ω_r :

$$\omega_r = \frac{1}{\sqrt{LC}}, \quad Z_{\text{total}}(\omega_r) = R.$$

The current at resonance is

$$I(\omega_r) = \frac{V_{\text{applied}}}{R}.$$

Voltage across the inductor:

$$V_L = IX_L = I \omega_r L.$$

Substituting I :

$$V_L = \frac{V_{\text{applied}}}{R} \omega_r L.$$

Therefore the quality factor (voltage ratio form) is

$$Q = \frac{V_L}{V_{\text{applied}}} = \frac{\omega_r L}{R}.$$

Using the capacitor voltage similarly,

$$V_C = IX_C = I \frac{1}{\omega_r C} \Rightarrow Q = \frac{V_C}{V_{\text{applied}}} = \frac{1}{\omega_r C R}.$$

Thus the two forms are equivalent:

$$\boxed{Q = \frac{\omega_r L}{R} = \frac{1}{\omega_r C R}}$$

Physical Note

At resonance the inductor and capacitor exchange energy, producing voltages across each that may be much larger than the applied voltage; the factor Q quantifies how many times larger these component voltages are compared to the supply.

Bandwidth of a Series Resonant Circuit and its Relation to Q

Consider a series RLC circuit with

$$Z(\omega) = R + j \left(\omega L - \frac{1}{\omega C} \right).$$

The magnitude of the impedance is

$$|Z(\omega)| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}.$$

At resonance ($\omega = \omega_0$) the reactive terms cancel:

$$\omega_0 L = \frac{1}{\omega_0 C} \quad \Rightarrow \quad \omega_0 = \frac{1}{\sqrt{LC}},$$

and the impedance is minimum:

$$Z_{\min} = |Z(\omega_0)| = R.$$

The current amplitude (for a fixed supply voltage V) is

$$I(\omega) = \frac{V}{|Z(\omega)|}, \quad I_{\max} = I(\omega_0) = \frac{V}{R}.$$

The half-power (-3 dB) frequencies ω_1 and ω_2 are defined by the condition that the power delivered to R is half its maximum value. Since $P \propto I^2 R$, the half-power condition is

$$I(\omega_{1,2}) = \frac{I_{\max}}{\sqrt{2}} \quad \Rightarrow \quad |Z(\omega_{1,2})| = \sqrt{2} R.$$

Therefore at $\omega_{1,2}$:

$$R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2 = 2R^2 \quad \Rightarrow \quad \left(\omega L - \frac{1}{\omega C}\right)^2 = R^2.$$

Thus

$$\omega L - \frac{1}{\omega C} = \pm R. \tag{1}$$

To solve (1) for $\omega_{1,2}$ we linearize the reactance difference about ω_0 . Define the small deviation $\Delta\omega = \omega - \omega_0$. For $|\Delta\omega| \ll \omega_0$, expand

$$\omega L - \frac{1}{\omega C} \approx (\omega - \omega_0) \left. \frac{d}{d\omega} \left(\omega L - \frac{1}{\omega C} \right) \right|_{\omega_0}.$$

Compute the derivative:

$$\frac{d}{d\omega} \left(\omega L - \frac{1}{\omega C} \right) = L + \frac{1}{\omega^2 C}.$$

At $\omega = \omega_0$, using $\omega_0 L = \frac{1}{\omega_0 C}$, we get

$$\left. \frac{d}{d\omega} \left(\omega L - \frac{1}{\omega C} \right) \right|_{\omega_0} = L + \frac{1}{\omega_0^2 C} = L + L = 2L.$$

Hence the linear approximation is

$$\omega L - \frac{1}{\omega C} \approx 2L(\omega - \omega_0) = 2L\Delta\omega.$$

Substitute into (1):

$$2L\Delta\omega = \pm R \quad \Rightarrow \quad \Delta\omega = \pm \frac{R}{2L}.$$

Therefore the half-power frequencies are

$$\omega_1 = \omega_0 - \frac{R}{2L}, \quad \omega_2 = \omega_0 + \frac{R}{2L}.$$

The (angular) *bandwidth* BW_ω is

$$BW_\omega = \omega_2 - \omega_1 = \frac{R}{L}.$$

Now recall the definition of the quality factor for a series RLC:

$$Q = \frac{\omega_0 L}{R}.$$

Rearranging gives $R/L = \omega_0/Q$. Substituting into the bandwidth expression yields the fundamental relation

$$BW_\omega = \frac{R}{L} = \frac{\omega_0}{Q}.$$

If bandwidth is expressed in hertz (cycles per second), use $\omega = 2\pi f$. Then

$$BW_f = f_2 - f_1 = \frac{BW_\omega}{2\pi} = \frac{\omega_0}{2\pi Q} = \frac{f_0}{Q},$$

so one commonly writes

$$\text{Bandwidth (Hz)} = \frac{f_0}{Q}.$$

Remarks

- The linearization step (expansion about ω_0) is valid when the circuit losses are small (R small) so that the half-power frequencies lie close to ω_0 (i.e. the circuit has moderately large Q).
- The same relations can be derived more exactly (without linearization) by solving the quadratic equation that follows from $(\omega L - 1/(\omega C))^2 = R^2$; the approximate result shown above is the common and useful form.

1. Circuit diagram o parallel resonance circuit

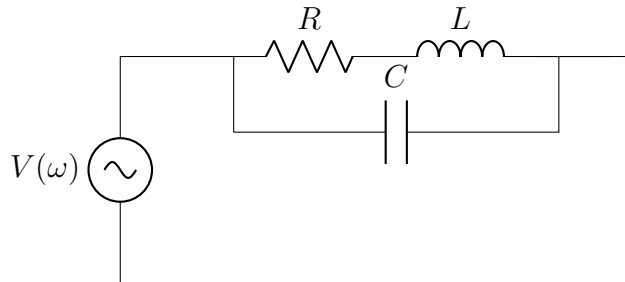


Figure 1: Series R and L branch in parallel with capacitor C .

2. Admittance and resonance condition

The series RL branch impedance:

$$Z_s(\omega) = R + j\omega L.$$

Its admittance:

$$Y_s(\omega) = \frac{1}{R + j\omega L} = \frac{R - j\omega L}{R^2 + \omega^2 L^2}.$$

Capacitor admittance:

$$Y_C(\omega) = j\omega C.$$

Total input admittance:

$$Y(\omega) = Y_s(\omega) + Y_C(\omega) = \frac{R}{R^2 + \omega^2 L^2} + j\left(\omega C - \frac{\omega L}{R^2 + \omega^2 L^2}\right).$$

Let the real part be $G(\omega)$ and imaginary part $B(\omega)$:

$$G(\omega) = \frac{R}{R^2 + \omega^2 L^2}, \quad B(\omega) = \omega C - \frac{\omega L}{R^2 + \omega^2 L^2}.$$

Resonance (anti-resonance) occurs when the susceptance vanishes:

$$B(\omega_0) = 0 \quad \implies \quad \omega_0 C = \frac{\omega_0 L}{R^2 + \omega_0^2 L^2}.$$

Cancelling ω_0 (assuming $\omega_0 \neq 0$) and rearranging:

$$C(R^2 + \omega_0^2 L^2) = L \quad \implies \quad \omega_0^2 = \frac{L - CR^2}{CL^2} = \frac{1}{LC} - \frac{R^2}{L^2}.$$

(is the resonance relation which we will use to find the input impedance of the circuit)

Hence the resonant angular frequency is

$$\omega_0 = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}.$$

Existence condition: the radicand must be positive:

$$\frac{1}{LC} - \frac{R^2}{L^2} > 0 \quad \implies \quad R < \sqrt{\frac{L}{C}}.$$

3. Input impedance at resonance

At ω_0 , $B(\omega_0) = 0$ so the admittance is purely real:

$$Y(\omega_0) = G(\omega_0) = \frac{R}{R^2 + \omega_0^2 L^2}.$$

Use $R^2 + \omega_0^2 L^2 = \frac{L}{C}$ (**from the resonance relation**) to get

$$Y(\omega_0) = \frac{R}{L/C} = \frac{RC}{L}.$$

Therefore the input impedance at resonance is

$$Z_{\text{in}}(\omega_0) = \frac{1}{Y(\omega_0)} = \frac{L}{RC}.$$

This is the maximum input impedance (anti-resonance). The circuit thus appears as a high impedance at ω_0 .

4. Quality factor Q

Ratio of capacitor current to total current is called as Quality Factor

Form the ratio:

$$\frac{I_C(\omega_0)}{I_{\text{total}}(\omega_0)} = \frac{\frac{V}{X_C(\omega_0)}}{\frac{V}{Z_{\text{in}}(\omega_0)}} = \frac{Z_{\text{in}}(\omega_0)}{X_C(\omega_0)}.$$

Using $Z_{\text{in}}(\omega_0) = \frac{L}{RC}$ and $X_C(\omega_0) = 1/(\omega_0 C)$ we obtain

$$\frac{I_C(\omega_0)}{I_{\text{total}}(\omega_0)} = \frac{L}{RC} \cdot \frac{1}{\frac{1}{\omega_0 C}} = Q.$$

But $\omega_0 = \frac{1}{\sqrt{LC}}$

$\therefore, Q = \frac{1}{R} \sqrt{\frac{L}{C}}$ A parallel resonating circuit is also frequently called rejector circuit or anti resonant circuit. Another name of the circuit is tuned circuit.

5. Summary (boxed results)

$$Y(\omega) = \frac{R}{R^2 + \omega^2 L^2} + j\left(\omega C - \frac{\omega L}{R^2 + \omega^2 L^2}\right)$$

$$\omega_0 = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}} \quad (\text{exists if } R < \sqrt{L/C})$$

$$Z_{\text{in}}(\omega_0) = \frac{L}{RC}, \quad Q = \frac{\omega_0 L}{R}$$

$$BW_{\omega} \approx \frac{\omega_0}{Q}, \quad BW_f \approx \frac{f_0}{Q}$$

Short Note on Active Power, Reactive Power, and Apparent Power

1. Active Power (Real Power), P

- **Definition:** It is the actual power consumed or utilized in an electrical circuit to perform useful work such as heating, lighting, or producing mechanical output.
- **Expression:**

$$P = VI \cos \phi = I^2 R = \frac{V^2}{R}$$

- **Unit:** Watt (W)
- **Remarks:** Active power represents the *useful power* converted into other forms of energy.

2. Reactive Power, Q

- **Definition:** Reactive power is the power that oscillates between the source and reactive components (inductors or capacitors) of the circuit. It does not perform useful work but is necessary for establishing magnetic and electric fields.
- **Expression:**

$$Q = VI \sin \phi = I^2 X = \frac{V^2}{X}$$

- **Unit:** Volt-Ampere Reactive (VAR)
- **Remarks:**
 - For inductive loads, reactive power is positive (lagging).
 - For capacitive loads, reactive power is negative (leading).

3. Apparent Power, S

- **Definition:** Apparent power is the vector sum of active and reactive powers. It represents the total power supplied by the source.
- **Expression:**

$$S = VI = \sqrt{P^2 + Q^2}$$

- **Unit:** Volt-Ampere (VA)
- **Remarks:** Apparent power indicates the total current and voltage requirement of the system, combining both useful and non-useful power components.

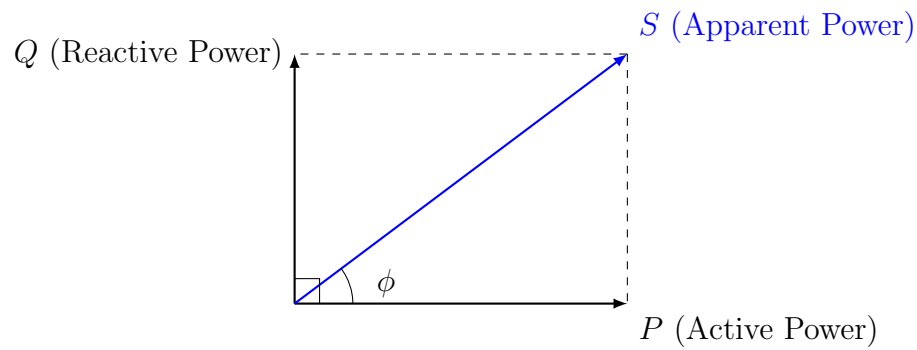
4. Relationship Between P , Q , and S

The three quantities are related by the **power triangle**:

$$S^2 = P^2 + Q^2$$

and

$$\cos \phi = \frac{P}{S} \quad (\text{Power Factor}).$$



Vector Relationship: $\vec{S} = \vec{P} + j\vec{Q}$

$S = P + jQ \quad \text{and} \quad S = \sqrt{P^2 + Q^2}$
--

5. Summary Table

Quantity	Symbol	Expression	Unit	Meaning
Active Power	P	$VI \cos \phi$	W	Useful (real) power consumed
Reactive Power	Q	$VI \sin \phi$	VAR	Power alternately stored and released
Apparent Power	S	VI	VA	Total supplied power

Short Note on Harmonics

Definition: Harmonics are sinusoidal components of a periodic waveform whose frequencies are integer multiples of the fundamental frequency f_1 . Thus the n -th harmonic has frequency nf_1 .

Key points:

- Any non-sinusoidal periodic waveform can be represented by a Fourier series as a sum of the fundamental and its harmonics.
- Harmonics are classified as *even* (2nd, 4th, ...) or *odd* (3rd, 5th, ...). Many practical sources produce predominantly odd harmonics.
- Harmonics cause waveform distortion, increased losses, overheating in machines and transformers, and power-quality problems.
- Control and mitigation methods include passive filters, active filters, proper converter design and power-factor correction.

EXAMPLE ON HARMONICS:

Given: A complex wave has total rms value $V_{\text{rms}} = 240$ V. The harmonic contents are:

3rd harmonic: 20% of fundamental, 5th harmonic: 5% of fundamental, 7th harmonic: 2% of fundamental

Let the fundamental RMS be V_1 . Then

$$V_3 = 0.20 V_1, \quad V_5 = 0.05 V_1, \quad V_7 = 0.02 V_1.$$

Because harmonics are orthogonal, the total RMS is the root-sum-square of the components:

$$V_{\text{rms}}^2 = V_1^2 + V_3^2 + V_5^2 + V_7^2 = V_1^2 (1 + 0.20^2 + 0.05^2 + 0.02^2).$$

Compute the multiplier:

$$1 + 0.20^2 + 0.05^2 + 0.02^2 = 1 + 0.04 + 0.0025 + 0.0004 = 1.0429.$$

Thus

$$V_1 = \frac{V_{\text{rms}}}{\sqrt{1.0429}} = \frac{240}{\sqrt{1.0429}} \approx 235.012 \text{ V}.$$

The harmonic RMS values are:

$$V_3 = 0.20 V_1 \approx 0.20 \times 235.012 \approx 47.002 \text{ V},$$

$$V_5 = 0.05 V_1 \approx 0.05 \times 235.012 \approx 11.751 \text{ V},$$

$$V_7 = 0.02 V_1 \approx 0.02 \times 235.012 \approx 4.700 \text{ V}.$$

Final results (rounded):

$V_1 \approx 235.01 \text{ V}, \quad V_3 \approx 47.00 \text{ V}, \quad V_5 \approx 11.75 \text{ V}, \quad V_7 \approx 4.70 \text{ V}$
--

Problem

An applied voltage

$$e(t) = 250 \sin(\omega t) + 50 \sin\left(3\omega t + \frac{\pi}{3}\right) + 20 \sin\left(5\omega t + \frac{5\pi}{6}\right)$$

is applied to a series circuit of $R = 20 \, \Omega$ and $L = 0.05 \, \text{H}$. Take $\omega = 314 \, \text{rad/s}$. Derive an expression for the current $i(t)$ (first find reactances X_1, X_3, X_5), find the RMS value of current and supply RMS voltage, the total real and reactive powers and the power factor. (No exponential notation used.)

1. Reactances

$$X_n = n\omega L.$$

Thus

$$X_1 = 1 \cdot 314 \cdot 0.05 = 15.700 \, \Omega,$$

$$X_3 = 3 \cdot 314 \cdot 0.05 = 47.100 \, \Omega,$$

$$X_5 = 5 \cdot 314 \cdot 0.05 = 78.500 \, \Omega.$$

2. Impedances (magnitude and angle)

For the n -th harmonic

$$Z_n = R + jX_n, \quad |Z_n| = \sqrt{R^2 + X_n^2}, \quad \angle Z_n = \tan^{-1}\left(\frac{X_n}{R}\right).$$

Numerically:

$$|Z_1| = \sqrt{20^2 + 15.700^2} \approx 25.426 \, \Omega, \quad \angle Z_1 \approx 38.132^\circ,$$

$$|Z_3| = \sqrt{20^2 + 47.100^2} \approx 51.170 \, \Omega, \quad \angle Z_3 \approx 66.993^\circ,$$

$$|Z_5| = \sqrt{20^2 + 78.500^2} \approx 81.008 \, \Omega, \quad \angle Z_5 \approx 75.706^\circ.$$

3. RMS voltages of harmonics

For a sinusoid $A \sin(n\omega t + \phi)$ the RMS magnitude is $V_n = A/\sqrt{2}$. Hence

$$V_1 = \frac{250}{\sqrt{2}} \approx 176.777 \, \text{V}, \quad V_3 = \frac{50}{\sqrt{2}} \approx 35.355 \, \text{V}, \quad V_5 = \frac{20}{\sqrt{2}} \approx 14.142 \, \text{V}.$$

4. RMS currents and phases

RMS current magnitude for each harmonic:

$$I_n = \frac{V_n}{|Z_n|}.$$

Compute numerically:

$$I_{1,\text{rms}} = \frac{176.777}{25.426} \approx 6.95255 \text{ A},$$

$$I_{3,\text{rms}} = \frac{35.355}{51.170} \approx 0.690933 \text{ A},$$

$$I_{5,\text{rms}} = \frac{14.142}{81.008} \approx 0.174578 \text{ A}.$$

Phase (sine-form) of each current harmonic: if the voltage harmonic is $A_n \sin(n\omega t + \phi_n)$ and $\angle Z_n$ is as above, then the current term is delayed by $\angle Z_n$. Thus the sine-form phase for i_n equals $\phi_n - \angle Z_n$. Numerically (convert ϕ_n to degrees: $\phi_1 = 0^\circ$, $\phi_3 = 60^\circ$, $\phi_5 = 150^\circ$):

$$\text{phase}_1 = 0^\circ - 38.1319^\circ \approx -38.1319^\circ,$$

$$\text{phase}_3 = 60^\circ - 66.9925^\circ \approx -6.9925^\circ,$$

$$\text{phase}_5 = 150^\circ - 75.7064^\circ \approx 74.2936^\circ.$$

Peak values (to use in the sine-form time expression):

$$I_{n,\text{peak}} = \sqrt{2} I_{n,\text{rms}}.$$

Thus

$$I_{1,\text{peak}} \approx \sqrt{2} \times 6.95255 \approx 9.83239 \text{ A},$$

$$I_{3,\text{peak}} \approx \sqrt{2} \times 0.690933 \approx 0.977127 \text{ A},$$

$$I_{5,\text{peak}} \approx \sqrt{2} \times 0.174578 \approx 0.246890 \text{ A}.$$

5. Current waveform (sine form)

Combine harmonic currents:

$$i(t) \approx 9.8324 \sin(\omega t - 38.1319^\circ) + 0.97713 \sin(3\omega t - 6.9925^\circ) + 0.24689 \sin(5\omega t + 74.2936^\circ) \text{ A}.$$

(Each coefficient is a peak amplitude; angles are the sine-form phases.)

6. Total RMS current and supply RMS voltage

Because harmonics are orthogonal,

$$I_{\text{rms}} = \sqrt{I_{1,\text{rms}}^2 + I_{3,\text{rms}}^2 + I_{5,\text{rms}}^2} \approx \sqrt{6.95255^2 + 0.690933^2 + 0.174578^2} \approx 6.98898 \text{ A}.$$

Total RMS supply voltage:

$$V_{\text{rms}} = \sqrt{V_1^2 + V_3^2 + V_5^2} \approx \sqrt{176.777^2 + 35.355^2 + 14.142^2} \approx 180.831 \text{ V}.$$

7. Real power (average), reactive power and apparent power

Real power per harmonic (since $P_n = I_n^2 R$):

$$P_1 = I_{1,\text{rms}}^2 R \approx (6.95255)^2 \times 20 \approx 966.759 \text{ W},$$

$$P_3 \approx (0.690933)^2 \times 20 \approx 9.5478 \text{ W},$$

$$P_5 \approx (0.174578)^2 \times 20 \approx 0.60955 \text{ W}.$$

Total real power

$$P \approx 966.759 + 9.5478 + 0.60955 \approx 976.916 \text{ W}.$$

Reactive power per harmonic (inductive):

$$Q_n = I_{n,\text{rms}}^2 (n\omega L).$$

Numerically:

$$Q_1 \approx (6.95255)^2 \times (1 \cdot 314 \cdot 0.05) \approx 758.906 \text{ var},$$

$$Q_3 \approx (0.690933)^2 \times (3 \cdot 314 \cdot 0.05) \approx 22.4850 \text{ var},$$

$$Q_5 \approx (0.174578)^2 \times (5 \cdot 314 \cdot 0.05) \approx 2.39247 \text{ var},$$

so

$$Q_{\text{total}} \approx 758.906 + 22.4850 + 2.39247 \approx 783.783 \text{ var} \quad (\text{lagging}).$$

Apparent power

$$S = V_{\text{rms}} I_{\text{rms}} \approx 180.831 \times 6.98898 \approx 1263.827 \text{ VA}.$$

8. Power factor

$$\text{PF} = \frac{P}{S} \approx \frac{976.916}{1263.827} \approx 0.77298 \quad (\text{lagging}).$$

$$i(t) \approx 9.8324 \sin(\omega t - 38.132^\circ) + 0.97713 \sin(3\omega t - 6.993^\circ) + 0.24689 \sin(5\omega t + 74.294^\circ) \text{ A},$$

$$I_{\text{rms}} \approx 6.9890 \text{ A}, \quad V_{\text{rms}} \approx 180.831 \text{ V},$$

$$P \approx 976.916 \text{ W}, \quad Q \approx 783.783 \text{ var}, \quad S \approx 1263.827 \text{ VA},$$

$$\text{PF} \approx 0.77298 \text{ (lagging)}.$$

1 Three phase system

A three-phase system consists of three alternating voltages (or currents) of equal magnitude and frequency, displaced by 120° from each other.

1. Phase Voltages

The instantaneous phase voltages of a balanced three-phase system are:

$$\begin{aligned}v_a(t) &= V_m \sin(\omega t), \\v_b(t) &= V_m \sin(\omega t - 120^\circ) = V_m \sin\left(\omega t - \frac{2\pi}{3}\right), \\v_c(t) &= V_m \sin(\omega t + 120^\circ) = V_m \sin\left(\omega t + \frac{2\pi}{3}\right).\end{aligned}$$

2. Phase Currents

For a balanced load, the phase currents are:

$$\begin{aligned}i_a(t) &= I_m \sin(\omega t - \phi), \\i_b(t) &= I_m \sin\left(\omega t - \frac{2\pi}{3} - \phi\right), \\i_c(t) &= I_m \sin\left(\omega t + \frac{2\pi}{3} - \phi\right).\end{aligned}$$

3. Line and Phase Voltages (Star Connection)

For a Y-connected system:

$$V_L = \sqrt{3} V_{ph}.$$

The line voltages are given by:

$$\begin{aligned}v_{ab}(t) &= \sqrt{3} V_m \sin(\omega t + 30^\circ), \\v_{bc}(t) &= \sqrt{3} V_m \sin(\omega t - 90^\circ), \\v_{ca}(t) &= \sqrt{3} V_m \sin(\omega t + 150^\circ).\end{aligned}$$

4. Line and Phase Currents (Delta Connection)

For a Δ -connected system:

$$I_L = \sqrt{3} I_{ph}.$$

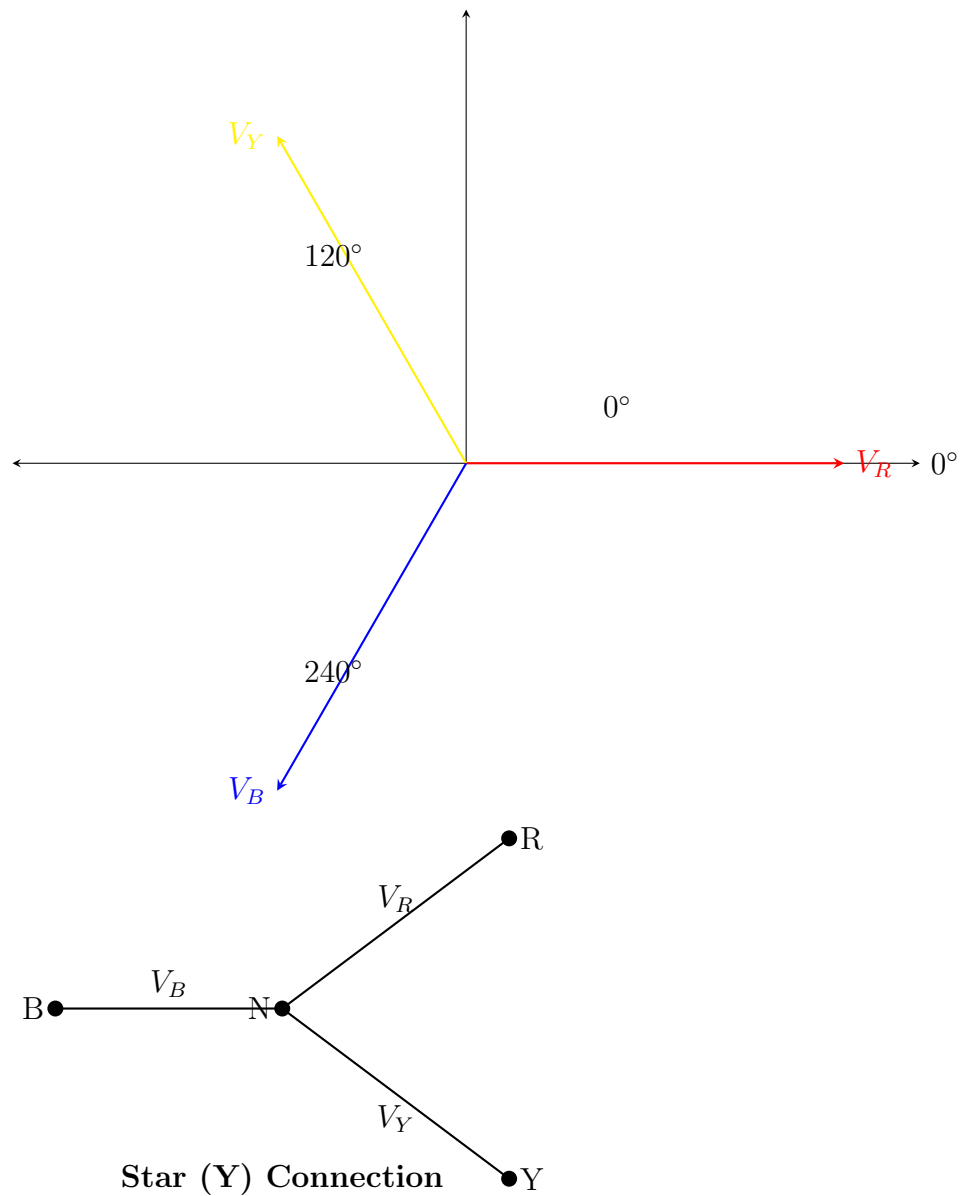
5. Total Power in a Three-Phase System

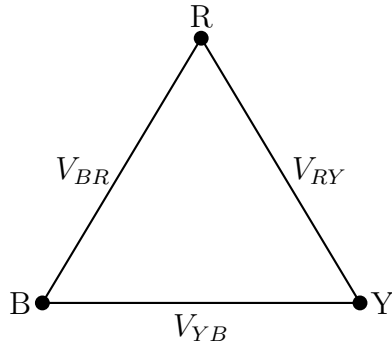
For a balanced load, the total active, reactive, and apparent powers are:

$$\begin{aligned}P &= \sqrt{3} V_L I_L \cos \phi, \\Q &= \sqrt{3} V_L I_L \sin \phi, \\S &= \sqrt{3} V_L I_L.\end{aligned}$$

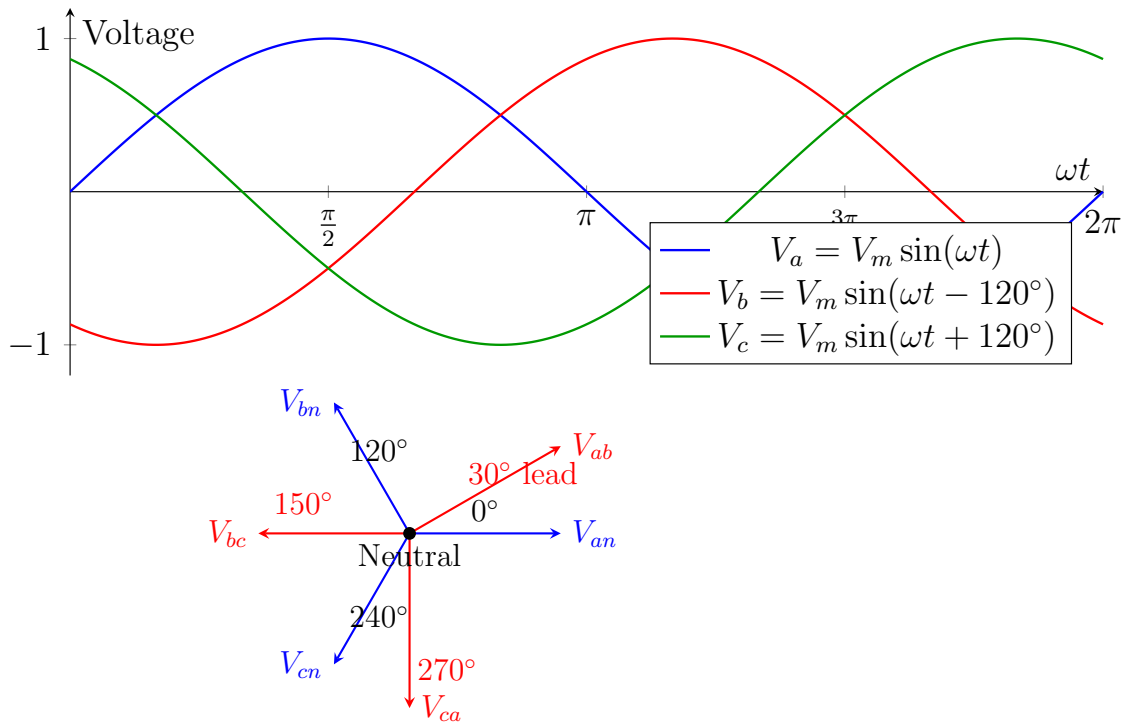
6. Advantages of Three-Phase System

- Constant power transfer.
- Higher efficiency in machines.
- Reduced conductor material.
- Smooth torque in motors.
- Produces a rotating magnetic field.





Delta (Δ) Connection



Line vs Phase Voltage Phasor Diagram (Y-System)

Proof: Line Voltage is $\sqrt{3}$ Times Phase Voltage

Consider a balanced three-phase star (Y) connected system. The phase voltages are:

$$\begin{aligned} V_{an} &= V_{ph} \angle 0^\circ, \\ V_{bn} &= V_{ph} \angle -120^\circ, \\ V_{cn} &= V_{ph} \angle 120^\circ. \end{aligned}$$

Line Voltage Expression

By definition,

$$V_{ab} = V_{an} - V_{bn}.$$

Substitute the phasors:

$$V_{ab} = V_{ph} \angle 0^\circ - V_{ph} \angle (-120^\circ) = V_{ph} (1 - e^{-j120^\circ}).$$

Convert to complex form:

$$e^{-j120^\circ} = \cos 120^\circ - j \sin 120^\circ = -\frac{1}{2} - j\frac{\sqrt{3}}{2}.$$

So,

$$1 - e^{-j120^\circ} = 1 + \frac{1}{2} + j\frac{\sqrt{3}}{2} = \frac{3}{2} + j\frac{\sqrt{3}}{2}.$$

Now compute the magnitude:

$$\begin{aligned} |V_{ab}| &= V_{ph} \sqrt{\left(\frac{3}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} \\ &= V_{ph} \sqrt{\frac{9}{4} + \frac{3}{4}} = V_{ph} \sqrt{\frac{12}{4}} = V_{ph} \sqrt{3}. \end{aligned}$$

Thus,

$$\boxed{V_L = \sqrt{3} V_{ph}}$$

Phase Difference

Also observe that:

$$\angle V_{ab} = 30^\circ.$$

So line voltage leads phase voltage by 30° .

Parallelogram Law Proof that $V_{RY} = \sqrt{3}V_R$

For a balanced three-phase system, the phase voltages are equal and displaced by 120° . Let V_R and V_Y be the phase voltages. The line voltage is

$$\mathbf{V}_{RY} = \mathbf{V}_R - \mathbf{V}_Y.$$

Using the vector magnitude formula,

$$|\mathbf{V}_{RY}| = \sqrt{V_R^2 + V_Y^2 - 2V_R V_Y \cos 120^\circ}.$$

Since $V_R = V_Y = V_{ph}$ and $\cos 120^\circ = -\frac{1}{2}$,

$$|\mathbf{V}_{RY}| = \sqrt{V_{ph}^2 + V_{ph}^2 - 2V_{ph}^2(-\frac{1}{2})} = \sqrt{3V_{ph}^2} = \sqrt{3} V_{ph}.$$

Thus,

$$\boxed{V_{RY} = \sqrt{3} V_R}.$$

Q1. What is a three-phase system? Explain its advantages.

Answer: A three-phase system consists of three alternating voltages of the same magnitude and frequency but displaced in time by 120° . If the phase voltages are represented as:

$$V_R = V_m \sin(\omega t), \quad V_Y = V_m \sin(\omega t - 120^\circ), \quad V_B = V_m \sin(\omega t - 240^\circ),$$

then the currents follow the same phase displacement (for a balanced load).

Advantages:

- More efficient transmission of power.
- Constant power transfer (unlike pulsating power in single-phase).
- Reduced conductor material for same power.
- Enables creation of a rotating magnetic field—foundation of AC motors.

Q2. What is the difference between phase voltage and line voltage?

Answer: In a star-connected system:

- **Phase voltage (V_{ph}):** Voltage across each phase winding.

$$V_{ph} = V_R = V_Y = V_B$$

- **Line voltage (V_L):** Voltage between any two line conductors.

$$V_L = V_{RY}, V_{YB}, V_{BR}$$

For a balanced star connection:

$$V_L = \sqrt{3} V_{ph}$$

In a delta connection:

$$V_L = V_{ph}$$

Q3. Explain why line voltage is $\sqrt{3}$ times phase voltage in a star connection.

Answer: Consider the line voltage between R and Y phases:

$$V_{RY} = V_R - V_Y$$

Representing the phase voltages as phasors separated by 120° , the resultant phasor has magnitude:

$$|V_{RY}| = \sqrt{V_R^2 + V_Y^2 - 2V_R V_Y \cos(120^\circ)}$$

Since $\cos(120^\circ) = -\frac{1}{2}$ and $V_R = V_Y = V_{ph}$,

$$|V_{RY}| = \sqrt{V_{ph}^2 + V_{ph}^2 + V_{ph}^2} = \sqrt{3} V_{ph}$$

Q4. What is a balanced three-phase load?

Answer: A three-phase load is said to be balanced if:

$$Z_R = Z_Y = Z_B = Z$$

and the three currents have:

$$I_R = I_m \sin(\omega t), \quad I_Y = I_m \sin(\omega t - 120^\circ), \quad I_B = I_m \sin(\omega t - 240^\circ)$$

In a balanced system:

$$I_R + I_Y + I_B = 0,$$

which implies that the neutral current is zero.

Q5. Explain the concept of power in a three-phase system.

Answer: The instantaneous power in each phase is:

$$p_R = V_R I_R, \quad p_Y = V_Y I_Y, \quad p_B = V_B I_B.$$

The total average power for a balanced system is:

$$P = \sqrt{3} V_L I_L \cos \phi.$$

Power is constant (non-pulsating), which is a major advantage of three-phase systems.

Q6. What is the phase sequence? Why is it important?

Answer: **Phase sequence** is the order in which the voltages reach their maximum positive values, typically R–Y–B.

Importance:

- Determines the direction of rotation of three-phase motors.
- Ensures proper operation of protective devices.
- Incorrect sequence can damage machinery.

Q7. What is a rotating magnetic field in a three-phase system?

Answer: When a balanced three-phase supply is applied to a three-phase stator winding, the three currents displaced by 120° create magnetic fluxes also displaced by 120° . Their vector sum produces a smoothly rotating magnetic field.

The speed of the rotating field:

$$N_s = \frac{120f}{P} \quad (\text{rpm})$$

where f is frequency and P is number of poles.

Q8. Compare star and delta connections.

Answer:

Star (Y) Connection:

- Neutral point exists.
- Line voltage is $\sqrt{3}$ times phase voltage.
- Used for long-distance transmission.

Delta Δ Connection:

- No neutral.
- Line current is $\sqrt{3}$ times phase current.
- Provides higher starting torque in motors.

Delta (Δ) Connection: Relation Between Line and Phase Current

In a delta-connected system, the three phase windings are connected end-to-end, forming a closed loop. Each line is connected at the junction of two phases. Therefore, the phase voltage is equal to the line voltage:

$$V_{ph} = V_L$$

Derivation: Line Current in (Δ) Connection

Consider the line current at terminal R:

$$I_R = I_{RY} - I_{BR}$$

The two phase currents meeting at node R are equal in magnitude but displaced by 120° .

Let the magnitude of phase current be I_{ph} .

The phasor difference is:

$$|I_R| = \sqrt{I_{ph}^2 + I_{ph}^2 - 2I_{ph}^2 \cos(120^\circ)}$$

Since

$$\cos(120^\circ) = -\frac{1}{2},$$

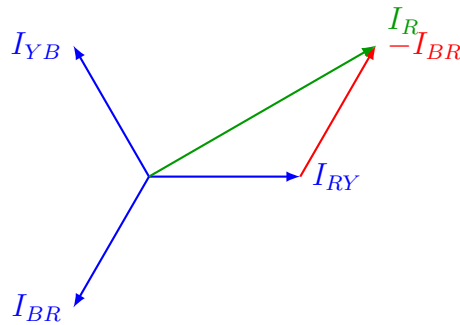
$$|I_R| = \sqrt{I_{ph}^2 + I_{ph}^2 + I_{ph}^2} = \sqrt{3} I_{ph}$$

Thus,

$$\boxed{I_L = \sqrt{3} I_{ph}}$$

Also, the line current leads the phase current by 30° .

Phasor Diagram Showing Line and Phase Currents in (Δ) Connection



Phasor Diagram of Phase Currents and Line Current in (Δ) Connection

Example 1: Power in a Balanced Star-Connected Load

A balanced three-phase star-connected load has a line voltage of $V_L = 415 \text{ V}$ and a line current of $I_L = 10 \text{ A}$. The power factor is $\cos \phi = 0.8$ lagging. Find the total power consumed.

Solution:

For a star connection,

$$P = \sqrt{3}V_L I_L \cos \phi.$$

Substituting the values,

$$P = \sqrt{3} \times 415 \times 10 \times 0.8$$

$$P = 1.732 \times 4150 \times 0.8 = 5749.76 \text{ W}$$

$$\boxed{P \approx 5.75 \text{ kW}}$$

Example 2: Line and Phase Currents in a Delta Load

A balanced delta-connected load has a phase impedance of

$$Z_{ph} = 20 + j10 \ \Omega.$$

The line voltage is

$$V_L = 400 \text{ V}.$$

Find the (a) phase current, (b) line current, and (c) power.

Solution:**(a) Phase current:**

In delta connection,

$$V_{ph} = V_L = 400 \text{ V}$$

$$I_{ph} = \frac{V_{ph}}{|Z_{ph}|} = \frac{400}{\sqrt{20^2 + 10^2}} = \frac{400}{22.36} = 17.9 \text{ A}$$

(b) Line current:

$$I_L = \sqrt{3}I_{ph} = 1.732 \times 17.9 = 31.0 \text{ A}$$

(c) Total power:

$$\cos \phi = \frac{20}{\sqrt{20^2 + 10^2}} = \frac{20}{22.36} = 0.895$$

$$P = \sqrt{3}V_L I_L \cos \phi$$

$$P = 1.732 \times 400 \times 31 \times 0.895$$

$$P = 19\,244 \text{ W}$$

$$\boxed{P \approx 19.2 \text{ kW}}$$

Example 3: Star Connection – Find Line Current

Three identical loads of

$$Z_{ph} = 15 \angle 30^\circ \Omega$$

are connected in star to a 440 V, 50 Hz supply. Find the line current.

Solution:

Phase voltage:

$$V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{440}{1.732} = 254 \text{ V}$$

Phase current:

$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{254}{15} = 16.93 \text{ A}$$

In a star load,

$$I_L = I_{ph} = 16.93 \text{ A}$$

$$\boxed{I_L = 16.9 \text{ A}}$$

Example 4: Three-Phase Power Given Resistance Load

A three-phase, 4-wire star system supplies a balanced resistive load of

$$R = 50 \, \Omega \text{ per phase.}$$

The line voltage is $V_L = 230 \text{ V}$. Find the line current and total power.

Solution:

Phase voltage:

$$V_{ph} = \frac{230}{\sqrt{3}} = 132.8 \text{ V}$$

Phase current:

$$I_{ph} = \frac{V_{ph}}{R} = \frac{132.8}{50} = 2.656 \text{ A}$$

For star load,

$$I_L = I_{ph} = 2.656 \text{ A}$$

Total power:

$$P = 3I_{ph}^2 R$$

$$P = 3 \times (2.656)^2 \times 50$$

$$P = 1059 \text{ W}$$

$$\boxed{P \approx 1.06 \text{ kW}}$$

Example 5: Power Measurement by Two-Wattmeter Method

A balanced 3-phase load draws a line current of

$$I_L = 20 \text{ A}$$

from a 440 V, 3-phase supply. The two wattmeters read:

$$W_1 = 5000 \text{ W}, \quad W_2 = 2000 \text{ W}$$

Find: (a) power factor, (b) total power.

Solution:

(a) Power factor:

$$\tan \phi = \frac{\sqrt{3}(W_1 - W_2)}{W_1 + W_2}$$
$$\tan \phi = \frac{\sqrt{3}(5000 - 2000)}{7000} = \frac{1.732 \times 3000}{7000} = 0.742$$
$$\phi = 36.6^\circ$$

$\cos \phi = 0.80$

(b) Total power:

$$P = W_1 + W_2 = 7000 \text{ W}$$

$P = 7 \text{ kW}$

Question–Answer: Wattmeter Internal Windings

Q1. What are the two internal windings of a wattmeter?

A1. A wattmeter consists of two internal windings:

- Current Coil (CC) or Series Coil
- Potential Coil (PC) or Pressure Coil

Q2. What is the function of the Current Coil (CC)?

A2. The Current Coil (CC) measures the load current. It is connected **in series** with the load and carries the full-load current.

$$I_{CC} = I_{\text{load}}$$

Q3. What are the electrical characteristics of the Current Coil?

A3.

- Very low resistance
- Made of thick wire
- Designed to carry load current

Its terminals are usually marked as:

$$M \text{ (main), } L \text{ (load)}$$

Q4. What is the function of the Potential Coil (PC)?

A4. The Potential Coil (PC) senses the voltage across the load. It is connected **in parallel** with the load.

$$V_{PC} = V_{load}$$

Q5. What are the electrical characteristics of the Potential Coil?

A5.

- Very high resistance
- Made of fine (thin) wire
- Only a small current flows through it

Its terminals are usually marked as:

$$V, \quad V'$$

Q6. How does a wattmeter measure power using its two coils?

A6. The wattmeter works on the principle of the interaction of magnetic fields produced by the two coils. The deflecting torque is proportional to:

$$\text{Torque} \propto I_{CC} \times V_{PC} \times \cos \phi$$

Thus, the wattmeter measures the true power:

$$P = VI \cos \phi$$

Q7. Why are polarity markings important in wattmeter coils?

A7. Correct polarity ensures proper phase relationship between voltage and current. If connections are reversed, the wattmeter may show:

- Negative reading
- Upscale deflection

Q1. Why do we use only two wattmeters to measure power in a three-phase system instead of three wattmeters? Give an example of a three-phase induction motor.

Answer: In a balanced three-phase system, the algebraic sum of instantaneous voltages is zero. Therefore, the total power of a three-phase load can be completely determined by using only **two wattmeters**. This is known as the **Two-Wattmeter Method**.

Using three wattmeters is theoretically possible, but it is:

- **Unnecessary** — because two wattmeters already provide the total power.
- **Not feasible in many cases** — especially when the load is connected in Δ (delta), where phase currents are not directly accessible.

Thus, the two-wattmeter method is simpler, economical, and accurate for balanced and unbalanced loads.

Example: Three-Phase Induction Motor

A three-phase induction motor is commonly connected in Δ . In this configuration:

- Only the **three line wires** enter the motor terminal box.
- Individual **phase currents cannot be measured** directly.
- Hence, placing a wattmeter in each phase (three-wattmeter method) becomes difficult.

However, the two-wattmeter method requires only:

Two current coils in two lines and Two voltage coils connected between lines

This makes it ideal for measuring:

- Power drawn by three-phase induction motors
- Power in industrial loads
- Balanced and unbalanced three-phase circuits

Three wattmeters are not practically feasible for loads like three-phase induction motors because individual phase access is not available. The two-wattmeter method remains the most convenient and widely used method for measuring total power in a three-phase system.

Two-Wattmeter Method — Formulae

For a three-phase system (line voltage V_L , line current I_L) supplying a load with power factor angle ϕ (lagging if inductive):

$$W_1 = V_L I_L \cos(\phi - 30^\circ),$$

$$W_2 = V_L I_L \cos(\phi + 30^\circ).$$

Useful derived relations:

$$P_{\text{total}} = W_1 + W_2 = \sqrt{3} V_L I_L \cos \phi,$$

$$W_1 - W_2 = V_L I_L \sin \phi,$$

$$\tan \phi = \sqrt{3} \frac{W_1 - W_2}{W_1 + W_2}.$$

Example 1 (Forward problem)

Given: Line voltage $V_L = 400\text{ V}$, line current $I_L = 10\text{ A}$, power factor $\cos \phi = 0.80$ (lagging).

Find: Wattmeter readings W_1 and W_2 , total power P .

Solution:

Compute the power factor angle:

$$\phi = \cos^{-1}(0.80) = 36.8699^\circ \quad (\text{approx}).$$

Now compute W_1 and W_2 :

$$W_1 = V_L I_L \cos(\phi - 30^\circ) = 400 \times 10 \times \cos(36.8699^\circ - 30^\circ)$$

$$= 4000 \times \cos(6.8699^\circ) \approx 4000 \times 0.993 \approx 3972\text{ W},$$

$$W_2 = V_L I_L \cos(\phi + 30^\circ) = 4000 \times \cos(66.8699^\circ) \approx 4000 \times 0.392 \approx 1568\text{ W}.$$

Total power (check):

$$P = W_1 + W_2 \approx 3972 + 1568 = 5540\text{ W}.$$

Using the standard three-phase formula:

$$P = \sqrt{3} V_L I_L \cos \phi = 1.732 \times 400 \times 10 \times 0.8 \approx 5542.4\text{ W},$$

(agreement within rounding).

Also the difference:

$$W_1 - W_2 \approx 3972 - 1568 = 2404\text{ W} \quad \text{and} \quad V_L I_L \sin \phi = 4000 \times \sin(36.8699^\circ) \approx 4000 \times 0.6 = 2400\text{ W},$$

(again close, rounding differences).

Example 2 (Inverse problem)

Given: Two wattmeter readings $W_1 = 5000\text{ W}$ and $W_2 = 2000\text{ W}$, line voltage $V_L = 440\text{ V}$. **Find:** (a) Total active power P , (b) power factor $\cos \phi$, (c) line current I_L .

Solution:

(a) Total power:

$$P = W_1 + W_2 = 5000 + 2000 = 7000\text{ W}.$$

(b) Power factor angle using the two-wattmeter relation:

$$\tan \phi = \sqrt{3} \frac{W_1 - W_2}{W_1 + W_2} = 1.732 \cdot \frac{5000 - 2000}{7000} = 1.732 \cdot \frac{3000}{7000} \approx 1.732 \times 0.42857 \approx 0.742.$$

Thus

$$\phi = \tan^{-1}(0.742) \approx 36.6^\circ, \quad \cos \phi \approx \frac{1}{\sqrt{1 + \tan^2 \phi}} \approx 0.80.$$

(c) Line current from $P = \sqrt{3} V_L I_L \cos \phi$:

$$I_L = \frac{P}{\sqrt{3} V_L \cos \phi} = \frac{7000}{1.732 \times 440 \times 0.80} \approx \frac{7000}{609.66} \approx 11.48\text{ A}.$$

Derivation of Two-Wattmeter Method

Consider a balanced three-phase system supplying a (possibly unbalanced) load. Let the line (phase) voltage magnitude be V_L and line current magnitude be I_L . Assume the line current $\mathbf{I}_a$ lags its corresponding phase voltage by an angle ϕ (i.e. a lagging power factor). We choose the phase a voltage reference such that

$$V_{an} = V_{ph} \angle 0^\circ.$$

For a balanced system the line (between lines) voltage V_{ab} is

$$\mathbf{V}_{ab} = \sqrt{3} V_{ph} \angle 30^\circ = V_L \angle 30^\circ.$$

The line current (phasor) is

$$\mathbf{I}_a = I_L \angle (-\phi).$$

A wattmeter with its potential coil connected across line a and line b (voltage V_{ab}) and its current coil in line a gives the reading (using RMS phasors and passive sign convention)

$$W_1 = \Re\{\mathbf{V}_{ab} \mathbf{I}_a^*\},$$

where $\Re\{\cdot\}$ denotes the real part and $*$ denotes complex conjugate.

Substitute the phasors:

$$\mathbf{V}_{ab} \mathbf{I}_a^* = (V_L \angle 30^\circ) (I_L \angle (+\phi)) = V_L I_L \angle (30^\circ + \phi).$$

Hence

$$\boxed{W_1 = V_L I_L \cos(30^\circ + \phi)}.$$

Similarly, the second wattmeter is usually connected with its potential coil across line c and line b (i.e. voltage V_{cb}) and its current coil in line c . For a balanced system,

$$\mathbf{V}_{cb} = V_L \angle (-90^\circ) \quad (\text{since } V_{cb} = V_c - V_b = \sqrt{3} V_{ph} \angle -90^\circ),$$

and

$$\mathbf{I}_c = I_L \angle (-120^\circ - \phi) \quad (\text{phasor position of } I_c).$$

Thus

$$\mathbf{V}_{cb} \mathbf{I}_c^* = V_L \angle (-90^\circ) I_L \angle (+120^\circ + \phi) = V_L I_L \angle (30^\circ - \phi).$$

So

$$\boxed{W_2 = V_L I_L \cos(30^\circ - \phi)}.$$

These two expressions are the usual pair of wattmeter readings (note that some texts swap the labels W_1, W_2 depending on the exact wiring/polarity; the algebra below is independent of that labelling).

Total active power

Add the two readings:

$$\begin{aligned} P_{\text{total}} &= W_1 + W_2 = V_L I_L [\cos(30^\circ + \phi) + \cos(30^\circ - \phi)] \\ &= 2V_L I_L \cos 30^\circ \cos \phi = 2V_L I_L \left(\frac{\sqrt{3}}{2} \right) \cos \phi \\ &= \sqrt{3} V_L I_L \cos \phi. \end{aligned}$$

Therefore

$$P = \sqrt{3} V_L I_L \cos \phi$$

which is the standard three-phase active power formula.

Difference of readings and power factor

The difference gives:

$$\begin{aligned} W_1 - W_2 &= V_L I_L [\cos(30^\circ + \phi) - \cos(30^\circ - \phi)] \\ &= -2V_L I_L \sin 30^\circ \sin \phi = -2V_L I_L \left(\frac{1}{2} \right) \sin \phi \\ &= -V_L I_L \sin \phi. \end{aligned}$$

Taking absolute values / appropriate sign convention, one commonly uses

$$|W_1 - W_2| = V_L I_L \sin \phi.$$

From the sum and difference one obtains the power factor angle:

$$\tan \phi = \sqrt{3} \frac{W_1 - W_2}{W_1 + W_2},$$

and therefore the power factor:

$$\cos \phi = \frac{P}{\sqrt{3} V_L I_L} = \frac{W_1 + W_2}{\sqrt{(W_1 + W_2)^2 + (\sqrt{3}(W_1 - W_2))^2}}.$$

Remark on sign convention: depending on the exact wiring and the definition of ϕ (one may define ϕ as current lag or lead, or choose different reference phasor), the individual wattmeter formulae can be written equivalently as

$$W_1 = V_L I_L \cos(\phi - 30^\circ), \quad W_2 = V_L I_L \cos(\phi + 30^\circ),$$

which is the same pair but with a different sign convention for ϕ . The important, invariant results are the *sum* $W_1 + W_2 = \sqrt{3} V_L I_L \cos \phi$ and the *tan* relation above.

Two wattmeters suffice to measure total active power in a three-phase system because their algebraic sum equals the total three-phase power; from their sum and difference one can also determine the power factor.

2 DC–DC Converters

A DC–DC converter is an electronic circuit used to convert one level of DC voltage into another level. Modern electrical and electronic systems contain multiple subsystems, each requiring a different DC voltage. A single DC source (such as a battery, solar panel, or DC bus) cannot satisfy all these voltage levels, and therefore DC–DC converters are required to step-up or step-down the voltage efficiently.

Unlike linear regulators, DC–DC converters offer:

- High efficiency (typically 85–95%)
- Regulated and stable output voltage
- Electrical isolation (in some topologies)
- Compact size with less heat dissipation

1. Different Loads Require Different Voltage Levels

Various subsystems in modern devices operate at different DC voltages. Examples are shown below:

Subsystem / Load	Typical Voltage
Digital logic ICs	1.0 V, 1.2 V, 1.8 V
Microcontrollers	3.3 V or 5 V
Sensors	2.5 V–3.3 V
Analog circuits	± 12 V or ± 15 V
Power amplifiers	24 V or 48 V
Battery-powered systems	3.7 V or 12 V input with multiple outputs

Since a single supply cannot provide all these voltage levels, DC–DC converters are necessary to obtain the required outputs.

2. Batteries and Renewable Sources Provide Unregulated DC

Examples:

- Li-ion battery: 3.0 V to 4.2 V
- Lead-acid battery: 10.5 V to 14.4 V
- Solar panel: 15 V to 22 V depending on sunlight
- EV battery: 48 V to 800 V systems

The voltage varies with load and environmental conditions. A DC–DC converter regulates this input and delivers a stable output voltage to the load.

3. Efficiency Requirements

Using a linear regulator to convert 12 V to 5 V wastes a large amount of power:

$$\eta = \frac{5}{12} \approx 41.6\%$$

A DC–DC buck converter, however, achieves efficiencies above 90%, improving battery life and reducing heat dissipation.

4. Isolation Requirement

Certain applications require electrical isolation between input and output for safety and noise reduction. Converters like the flyback, forward, and push-pull topologies provide galvanic isolation. These are commonly used in:

- SMPS chargers
- Industrial control systems
- Medical equipment

Applications of DC–DC Converters

- Digital circuits requiring 1.0 V, 1.2 V, 1.8 V, 3.3 V, or 5 V
- Communication circuits such as RF amplifiers and PLLs
- Electric vehicles (EV): 400–800 V battery converted to 12 V auxiliary supply
- Computers and servers with multiple rails: 12 V, 5 V, 3.3 V, 1.8 V, 1.2 V
- Portable devices such as smartphones
- LED drivers and SMPS systems

Types of DC–DC Converters

- Buck converter (step-down)
- Boost converter (step-up)
- Buck–Boost converter (step-up/step-down)
- Flyback converter (isolated)
- SEPIC and Çuk converters (low-ripple step-up/step-down)

Summary

DC–DC converters are essential in modern electronic systems because:

- Different loads require different voltage levels.
- Input sources provide unregulated and varying DC.
- High efficiency and compact power management are required.
- Isolation is needed in many sensitive applications.

Thus, voltages such as 1.8 V, 3.3 V, 5 V, 12 V, 24 V, and 48 V are efficiently obtained from a single DC source using DC–DC converters.

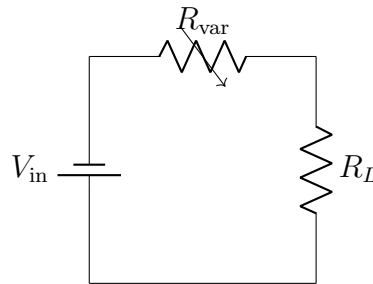
Voltage Control Using Series Variable Resistor and Its Drawbacks

A simple method to obtain different voltage levels from a DC supply is to connect a variable resistor (R_{var}) in series with the load resistor (R_L). By adjusting the value of the series resistor, the voltage drop across the load changes according to the voltage divider principle.

$$V_L = V_{\text{in}} \left(\frac{R_L}{R_{\text{var}} + R_L} \right)$$

Thus, various output voltages can be obtained simply by changing R_{var} .

Circuit Diagram



Explanation

When current flows through the series combination of R_{var} and R_L , the input voltage divides between them. The voltage across the load is:

$$V_L = IR_L,$$

where the current is:

$$I = \frac{V_{\text{in}}}{R_{\text{var}} + R_L}.$$

By increasing R_{var} , the current decreases, hence the load voltage decreases. Conversely, reducing R_{var} increases the load voltage. This is the simplest method of voltage control.

Drawbacks of This Method

Although this method works, it has significant disadvantages:

1. Power Dissipation in the Series Resistor

The power dissipated in the variable resistor is:

$$P_{\text{var}} = I^2 R_{\text{var}}.$$

Since R_{var} may be large for obtaining lower voltages, the power loss becomes substantial. This energy is wasted as heat.

2. Low Efficiency

The efficiency is:

$$\eta = \frac{P_L}{P_{\text{in}}} = \frac{I^2 R_L}{I^2 (R_{\text{var}} + R_L)} = \frac{R_L}{R_{\text{var}} + R_L}.$$

As R_{var} increases, efficiency decreases rapidly.

3. Load Voltage Depends on Load Current

If R_L changes, the output voltage V_L also changes:

$$V_L = V_{\text{in}} \left(\frac{R_L}{R_{\text{var}} + R_L} \right)$$

This means the circuit cannot maintain a regulated output voltage.

4. Thermal Problems

Large resistors may heat up, causing:

- drift in resistance value,
- reduced reliability,
- need for bulky high-power resistors.

5. Not Suitable for Sensitive Electronics

Since voltage varies with load, it cannot be used to power digital circuits, microcontrollers, sensors, or communication circuits that require precise stable DC supply.

Using a series variable resistor for voltage control is simple but highly inefficient and impractical for modern electronics. The drawbacks of heat generation, poor voltage regulation, and low efficiency make it unsuitable. Therefore, regulated and efficient DC–DC converters are preferred for obtaining stable low-voltage outputs.

section*Eliminating Power Dissipation in a Series Control Element When a variable resistor is used in series with a load to control the output voltage, the power dissipated

across the resistor is

$$P = I^2 R.$$

There are two possible ways to make this power dissipation zero:

1. **Make the current zero.** If the current through the series element is zero, then

$$P = I^2 R = 0,$$

regardless of the value of resistance. In this condition, no power is delivered to the load.

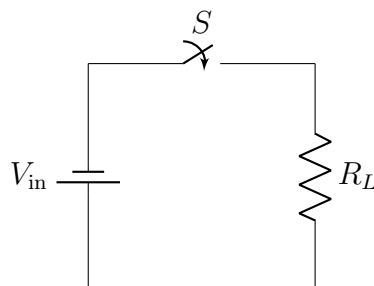
2. **Make the series resistance zero.** If the series resistor is replaced by an element whose effective resistance is zero,

$$P = I^2 \times 0 = 0,$$

meaning no power is dissipated in the controlling element.

To achieve this practically, instead of using a resistor we replace it with a **switch**. A switch in its ON state has nearly zero resistance, and in its OFF state it allows zero current. Thus, it satisfies both the above conditions. By rapidly switching the supply ON and OFF (using a suitable duty cycle), a **pulsating DC waveform** is produced, whose average value is controlled.

$$V_{\text{avg}} = D V_{\text{in}}, \quad \text{where } D = \text{duty cycle}$$

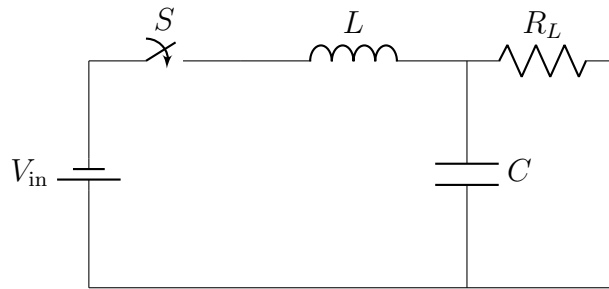


This method provides a **variable DC output voltage** with **nearly zero power loss** in the series control element. Therefore, the overall efficiency is much higher than using a series variable resistor. This is the basic operating principle of DC–DC converters using switching devices.

When a switch is used in place of a variable resistor, the output voltage becomes pulsating DC. This waveform contains:

- a DC component (average value),
- AC components (ripple), mostly at the switching frequency and its harmonics.

Load devices—especially DC motors, digital circuits, and battery chargers—usually require a smooth DC voltage. To remove these oscillations, an LC filter is used at the output.



Why an LC filter?

Inductor (L) resists sudden changes in current.

It smooths the current by blocking high-frequency components.

Capacitor (C) resists sudden changes in voltage.

It smooths the voltage by shunting high-frequency components to ground.

When used together:

L filters the ripple from the current side

C filters the ripple from the voltage side

Thus, ripple is reduced much more effectively than with a single L or C alone.

Primary Topologies in DC–DC Converters

In switch-mode DC–DC converters, when the circuit uses only **one switch**, **one inductor**, and **one capacitor**, the converter is classified as a **primary topology**. Such converters are the simplest forms of power electronic converters and operate by controlling the duty cycle of the switching device.

There are three basic (primary) topologies:

1. Step-Down (Buck) Converter

A buck converter produces an output voltage lower than the input voltage:

$$V_{\text{out}} < V_{\text{in}}.$$

The inductor smooths the pulsating current from the switch, and the capacitor reduces the output voltage ripple. Buck converters are widely used in digital electronics where regulated voltages such as 5 V, 3.3 V, or 1.8 V are required.

2. Step-Up (Boost) Converter

A boost converter produces an output voltage higher than the input voltage:

$$V_{\text{out}} > V_{\text{in}}.$$

During the ON period of the switch, the inductor stores energy, and during the OFF period it releases energy to the load at a higher voltage. Boost converters are used in LED drivers, battery charging circuits, and power-factor correction systems.

3. Step-Up/Step-Down (Buck–Boost) Converter

A buck–boost converter can produce an output voltage either higher or lower than the input voltage:

$$V_{\text{out}} \gtrless V_{\text{in}}.$$

This is achieved by inverting or rearranging the energy transfer path between the switch, inductor, and capacitor. Buck–boost converters are commonly used in battery-powered systems where the battery voltage varies significantly during operation.

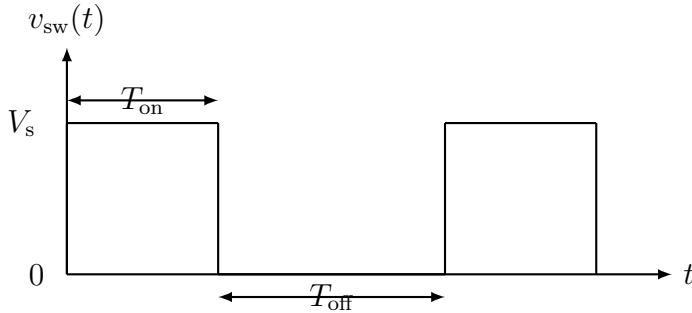
Summary

Topology	Output Level	Components	Applications
Buck	$V_{\text{out}} < V_{\text{in}}$	1 switch, L, C	Digital electronics
Boost	$V_{\text{out}} > V_{\text{in}}$	1 switch, L, C	LED drivers, chargers
Buck–Boost	$V_{\text{out}} \gtrless V_{\text{in}}$	1 switch, L, C	Battery regulators

Use of SPDT Switch in DC–DC Converter

In many DC–DC converter circuits, an SPST (Single-Pole Single-Throw) switch is used to connect or disconnect the inductor from the supply. However, the SPST switch has a disadvantage: when the switch is in the OFF state, the inductor input node becomes floating. A floating node may lead to undefined voltage levels, switching noise, or unstable behaviour of the converter.

To avoid this issue, an SPDT (Single-Pole Double-Throw) switch is used instead. One throw of the SPDT switch is connected to the supply voltage, while the other throw is connected to ground. The common terminal (pole) of the switch is connected to the LC filter input. This ensures that the LC input node is always referenced either to the supply or to ground, eliminating the floating condition and providing stable and predictable switching behaviour.



Concept Explanation: Negative Inductor Voltage During OFF State

When the SPDT switch is in the ON position, the inductor is connected to the supply voltage and the voltage across the inductor is

$$v_L = V_s - V_o > 0.$$

During this interval the inductor current increases with positive slope.

When the switch is moved to the OFF position, the inductor is connected to ground through the second throw of the switch. However, the inductor current cannot change instantaneously. To maintain the continuity of current, the inductor forces the current to keep flowing in the same direction.

Since the supply is disconnected in the OFF interval, the only way to maintain the current is for the inductor to reverse its polarity. This causes the voltage across the inductor to become *negative*:

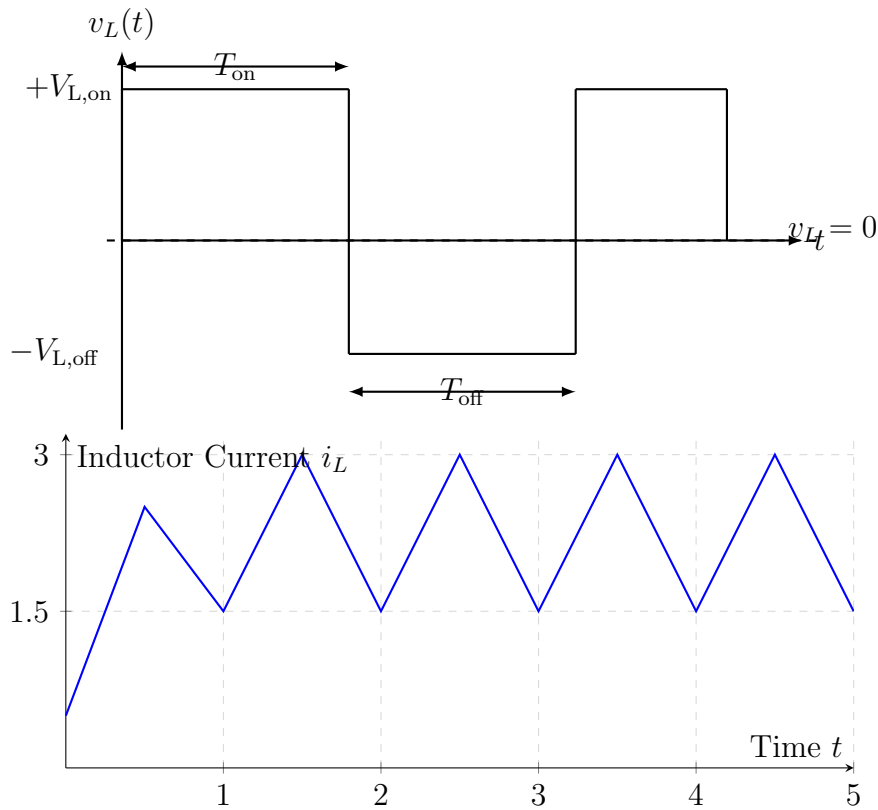
$$v_L = -V_o \quad (\text{OFF interval})$$

Thus, during OFF time the inductor voltage becomes negative and the inductor current decreases with a negative slope.

In steady-state operation of a DC-DC converter, the average inductor voltage must be zero:

$$\overline{v_L(t)} = 0.$$

Inductor Voltage Waveform



In the steady state conditions, average voltage across the inductor should be zero. This will avoid the inductor to go into saturation by building the flux. There should be a volt-second balance in case of the inductor also in case of capacitor, in the steady state condition, the average current should be zero otherwise there will be a charge developed. So, current-second balance is required in case of capacitor.

Question:

Explain the **volt-second balance** and **ampere-second balance** in the case of a DC-DC converter. Derive the expressions for a step-down (buck) converter.

Answer:

1. Volt-Second Balance

Definition: The *volt-second balance principle* states that, in steady state, the average voltage across an inductor over one switching period is zero:

$$\overline{v_L} = \frac{1}{T_s} \int_0^{T_s} v_L(t) dt = 0$$

For a **buck converter**:

- When the switch is ON ($0 \leq t \leq DT_s$):

$$v_L = V_{in} - V_o$$

- When the switch is OFF ($DT_s \leq t \leq T_s$):

$$v_L = -V_o$$

Applying volt-second balance:

$$(V_{in} - V_o) \cdot DT_s + (-V_o) \cdot (1 - D)T_s = 0$$

$$V_{in}D - V_o = 0 \implies V_o = D \cdot V_{in}$$

This is the **output voltage formula** of the buck converter.

2. Ampere-Second Balance

Definition: The *ampere-second balance principle* states that, in steady state, the average current through the output capacitor over one switching period is zero:

$$\overline{i_C} = \frac{1}{T_s} \int_0^{T_s} i_C(t) dt = 0$$

For the buck converter:

$$i_C = i_L - i_o$$

Applying ampere-second balance over one period:

$$\int_0^{T_s} (i_L - i_o) dt = 0 \implies \overline{i_L} = i_o$$

Interpretation: The average inductor current equals the load current.

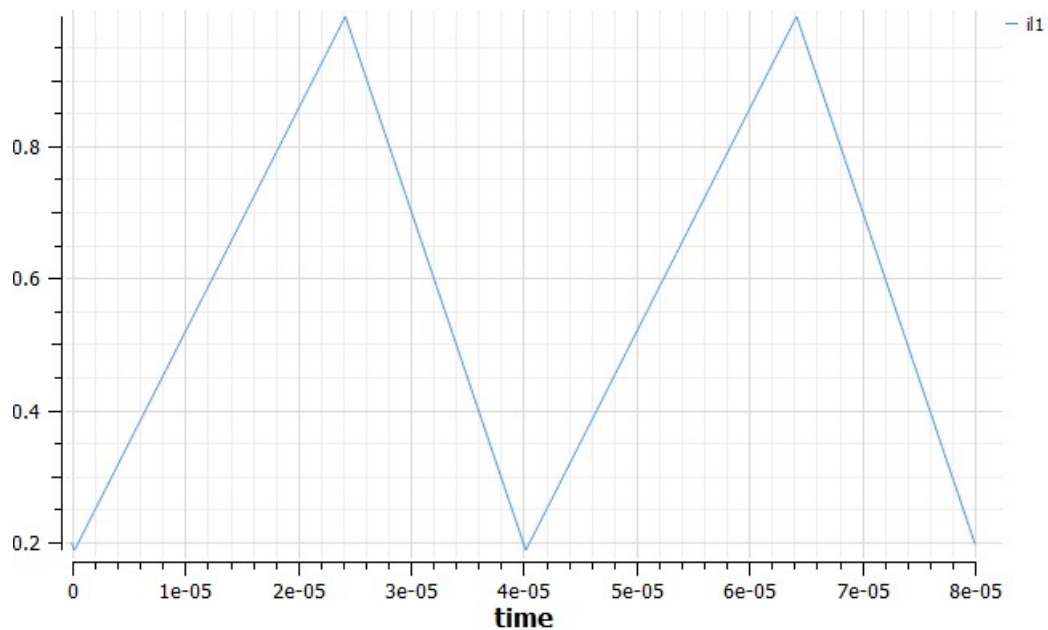
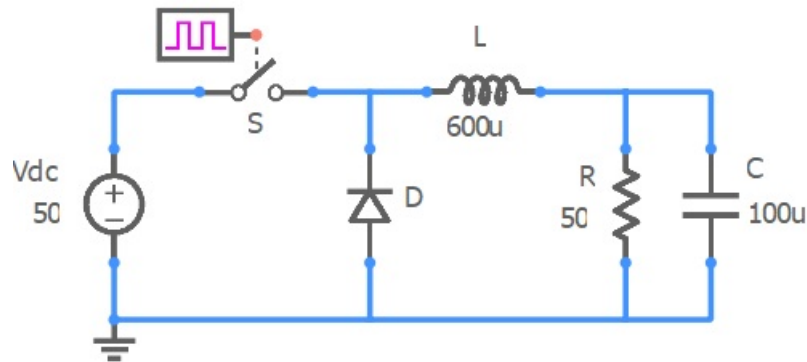
Summary

Principle	What it Ensures	Buck Converter Formula
Volt-Second Balance	Average inductor voltage = 0	$V_o = D \cdot V_{in}$
Ampere-Second Balance	Average capacitor current = 0	$\overline{i_L} = i_o$

Summary Table

Principle	What it Ensures	Buck Converter Formula
Volt-Second Balance	Average inductor voltage = 0	$V_o = D \cdot V_{in}$
Ampere-Second Balance	Average capacitor current = 0	$\bar{i}_L = i_o$

Voltage second equation will be as follows.



Why the Average Inductor Voltage and Average Capacitor Current Must Be Zero

1. Average Voltage Across the Inductor Must Be Zero

For an inductor, the fundamental relation is:

$$V_L = L \frac{di_L}{dt}$$

In steady-state, the inductor current must return to the same value at the end of each switching period. Therefore, the net change in inductor current over one switching cycle is:

$$\Delta i_L = \int_0^{T_s} \frac{di_L}{dt} dt = 0$$

This implies:

$$\int_0^{T_s} V_L(t) dt = 0$$

Or equivalently, the average inductor voltage is zero:

$$\overline{V_L} = \frac{1}{T_s} \int_0^{T_s} V_L(t) dt = 0$$

Expanding for ON and OFF intervals:

$$\int_0^{T_{\text{on}}} V_L dt + \int_{T_{\text{on}}}^{T_s} V_L dt = 0$$

If this condition is violated, the inductor current will drift upward or downward each cycle. In particular, if

$$\overline{V_L} > 0,$$

then the average current will increase, possibly driving the inductor into saturation and damaging the converter.

2. Average Current Through the Capacitor Must Be Zero

For a capacitor, the governing relation is:

$$i_C = C \frac{dv_C}{dt}$$

To maintain steady-state operation, the capacitor voltage must return to the same value at the end of each switching period. Thus the net change in capacitor voltage over one period is:

$$\Delta v_C = \int_0^{T_s} \frac{dv_C}{dt} dt = 0$$

Which implies:

$$\int_0^{T_s} i_C(t) dt = 0$$

Hence, the average capacitor current must be zero:

$$\overline{i_C} = \frac{1}{T_s} \int_0^{T_s} i_C(t) dt = 0$$

If the average capacitor current were not zero, the capacitor voltage would continuously increase or decrease, preventing steady-state and causing output voltage instability.

Summary

- The **inductor volt-second balance** condition:

$$\overline{V_L} = 0$$

ensures that inductor current does not drift or saturate.

- The **capacitor charge balance** condition:

$$\overline{i_C} = 0$$

ensures that capacitor voltage remains stable.

These two conditions are fundamental in the steady-state analysis of all DC–DC converters.

Assumptions for Analyzing a DC–DC Buck Converter

While analyzing the steady-state operation of a DC–DC buck converter, several simplifying assumptions are made. These assumptions allow closed-form expressions for inductor current, output voltage, and converter efficiency. The commonly used assumptions are:

1. **All components are ideal.** The switch has zero on-resistance and infinite off-resistance. The diode has zero forward voltage drop. The inductor and capacitor are ideal, meaning they have no internal losses:

$$r_L = 0, \quad ESR_C = 0.$$

2. **The output capacitor is sufficiently large.** This ensures that the output voltage ripple is very small, so:

$$v_o(t) \approx V_o \quad (\text{almost constant}).$$

3. **The inductor current ripple is small.** A large inductance ensures nearly constant current:

$$i_L(t) \approx I_o \quad \text{with small ripple.}$$

This is the Continuous Conduction Mode (CCM) assumption.

4. **For a buck converter:** The average output voltage is always less than the input voltage:

$$V_o = DV_{\text{in}}, \quad V_o < V_{\text{in}}.$$

5. **The load is purely resistive.** Therefore, the output current is given by Ohm's law:

$$I_o = \frac{V_o}{R}.$$

These assumptions simplify the analysis and help derive the fundamental buck converter relationships, such as:

$$V_o = DV_{in}, \quad I_{in} = DI_o.$$

$$\begin{aligned} \text{Area of the triangle} &= \frac{1}{2} * \frac{T_s}{2} * \frac{\Delta i_L}{2} = Q_0 \\ Q_0 &= \frac{1}{8} * \frac{V_{dc}(1-D)D}{L} * T_s^2 \\ Q_0 &= C\Delta V \end{aligned}$$

$$\text{ripple voltage} = \Delta V_0 = \frac{Q_0}{C} = \frac{1}{8} * \frac{V_{dc}(1-D)D}{LC} * T_s^2$$

Output ripple is inversely proportional to F_s^2 and also inversely proportional to the product of L and C. Hence to reduce the output voltage ripple either increase switching frequency or LC product.

Difference Between L_{critical} and L_{ripple} in a Buck Converter

1. Critical Inductance L_{critical}

Definition: L_{critical} is the minimum inductance required to keep the buck converter operating in **Continuous Conduction Mode (CCM)**.

$$L_{\text{critical}} = \frac{(1-D)R}{2f_s}$$

where

- D = duty cycle
- R = load resistance
- f_s = switching frequency

Meaning:

$$L < L_{\text{critical}} \Rightarrow \text{DCM (inductor current reaches zero)}$$

$$L \geq L_{\text{critical}} \Rightarrow \text{CCM}$$

2. Ripple-Based Inductance L_{ripple}

Definition: L_{ripple} is the inductance required to limit the inductor current ripple Δi_L to a specified value.

The ripple expression is:

$$\Delta i_L = \frac{(V_{in} - V_o)D}{Lf_s}$$

Rearranging for the inductance,

$$L_{\text{ripple}} = \frac{(V_{in} - V_o)D}{\Delta i_L f_s}$$

Meaning:

- Larger $L_{\text{ripple}} \rightarrow$ smaller ripple
- Smaller $L_{\text{ripple}} \rightarrow$ larger ripple

3. Key Differences

Parameter	L_{critical}	L_{ripple}
Purpose	Ensures CCM operation	Limits ripple current
Depends on	R, D, f_s	$V_{in}, V_o, \Delta i_L$
Related to	Mode boundary (CCM/DCM)	Performance (ripple)
Typical Magnitude	Smaller	Larger
Impact if too small	Enters DCM	Large current ripple

4. Why Inductance Must Be Larger Than Both?

(a) To Maintain CCM

$$L > L_{\text{critical}} \Rightarrow \text{Inductor current never reaches zero.}$$

(b) To Limit Current Ripple

If $L < L_{\text{ripple}}$:

- Ripple current becomes large
- Higher conduction losses
- Increased output voltage ripple
- Inductor may saturate

Thus,

$$L_{\text{selected}} > \max(L_{\text{critical}}, L_{\text{ripple}})$$

$$L_{\text{selected}} > \max(L_{\text{critical}}, L_{\text{ripple}})$$

This ensures:

- Stable CCM operation
- Acceptable ripple current

- Better efficiency
- Reduced stress on components

Q. Derive the expression for the output voltage of a DC–DC boost converter in terms of duty cycle D and input voltage V_{in} .

Answer:

A boost converter increases the input DC voltage to a higher DC output voltage. Let the switch be ON for time DT_s and OFF for $(1 - D)T_s$, where D is the duty cycle and T_s is the switching period.

During the ON interval, the inductor stores energy from the input source:

$$V_L = V_{in} \quad \text{for } DT_s$$

During the OFF interval, the switch opens, diode conducts, and the inductor releases energy to the output:

$$V_L = V_{in} - V_0 \quad \text{for } (1 - D)T_s$$

For steady-state continuous conduction mode (CCM), the average inductor voltage over one switching period must be zero (volt-second balance):

$$V_{in}(DT_s) + (V_{in} - V_0)((1 - D)T_s) = 0$$

Divide both sides by T_s :

$$V_{in}D + (V_{in} - V_0)(1 - D) = 0$$

Expand:

$$V_{in}D + V_{in} - V_{in}D - V_0 + DV_0 = 0$$

Simplify:

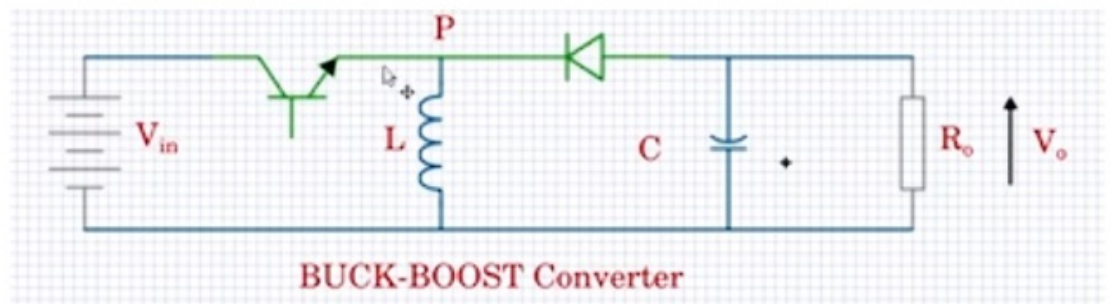
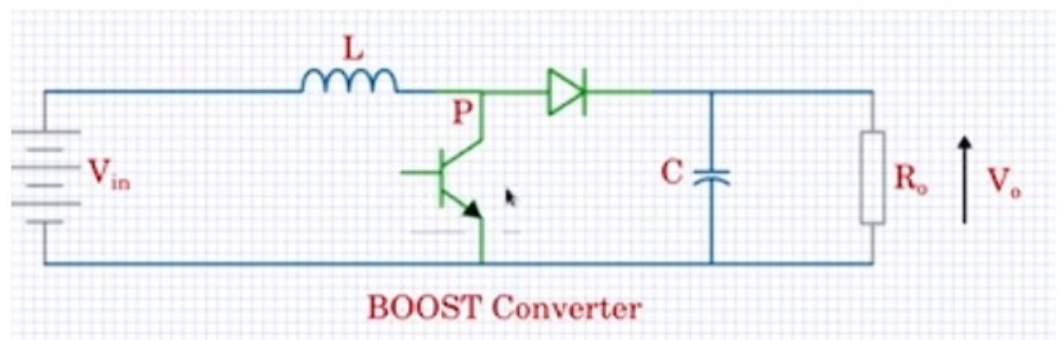
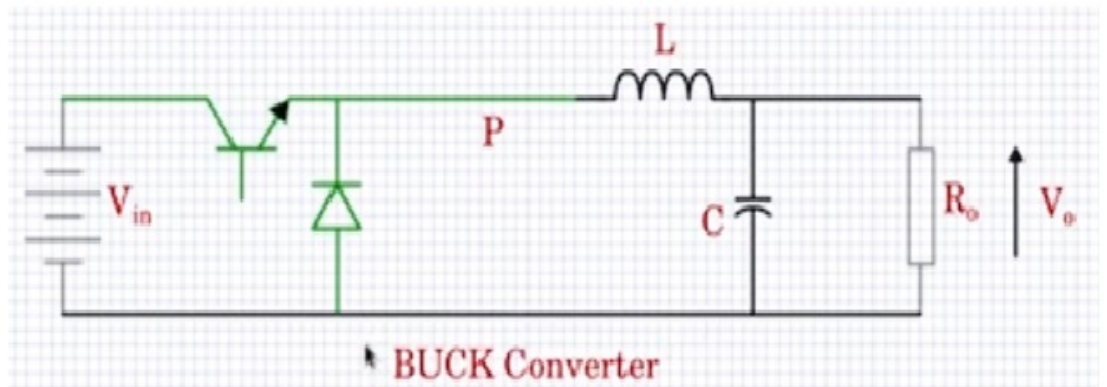
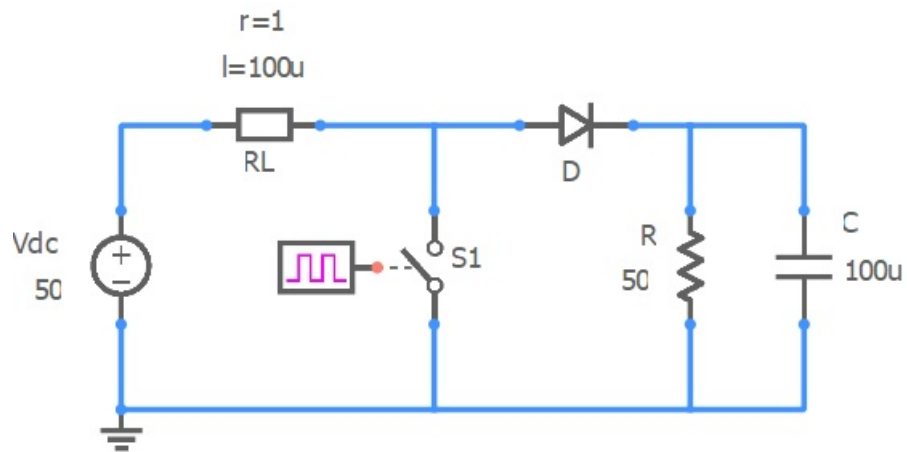
$$V_{in} - V_0(1 - D) = 0$$

$$V_0(1 - D) = V_{in}$$

Solving for V_0 :

$$\boxed{V_0 = \frac{V_{in}}{1 - D}}$$

Thus, the boost converter output voltage is greater than the input and increases as the duty cycle D approaches 1.



$$d = \text{dutytime} = \frac{T_{ON}}{T_{ON} + T_{OFF}}$$

Numerical :

Problem on DC–DC Boost Converter

Question:

A DC–DC boost converter operates in continuous conduction mode (CCM) with ideal switches at a switching frequency of 50 kHz. It is designed to regulate an output voltage of $V_o = 500$ V at an output power of $P_o = 500$ W. The applied DC input voltage varies in the range $100 \text{ V} \leq V_{\text{in}} \leq 250 \text{ V}$. The maximum allowable inductor current ripple is 20% of the output current, and the maximum allowable output voltage ripple is 2% of the output voltage.

Determine:

1. The average values of the switch current and diode current.
2. The minimum inductance and minimum output capacitance.
3. The duty cycle range of the converter over the input voltage range.

Solution

Given:

$$P_o = 500 \text{ W}, \quad V_o = 500 \text{ V}$$

$$f_s = 50 \text{ kHz}, \quad T_s = \frac{1}{f_s} = 20 \mu\text{s}$$

$$100 \text{ V} \leq V_{\text{in}} \leq 250 \text{ V}$$

$$\Delta I_L = 0.2 I_o, \quad \Delta V_o = 0.02 V_o$$

$$I_o = \frac{P_o}{V_o} = \frac{500}{500} = 1 \text{ A}$$

$$\Delta I_L = 0.2 I_o = 0.2(1) = 0.2 \text{ A}$$

$$\Delta V_o = 0.02 V_o = 0.02(500) = 10 \text{ V}$$

—

1. Duty Cycle Range

For a boost converter,

$$V_o = \frac{V_{\text{in}}}{1 - D}$$

$$D = 1 - \frac{V_{\text{in}}}{V_o}$$

At minimum input voltage $V_{\text{in}} = 100 \text{ V}$

$$D_{\text{max}} = 1 - \frac{100}{500} = 0.8$$

At maximum input voltage $V_{\text{in}} = 250 \text{ V}$

$$D_{\text{min}} = 1 - \frac{250}{500} = 0.5$$

$$\boxed{D \in [0.5, 0.8]}$$

—

2. Average Switch and Diode Currents

Inductor average current:

$$I_L = \frac{I_o}{1 - D}$$

The worst case (maximum current) occurs at $D_{\max} = 0.8$:

$$I_{L,\max} = \frac{1}{1 - 0.8} = \frac{1}{0.2} = 5 \text{ A}$$

Switch average current

Switch conducts for the interval DT_s :

$$I_{\text{sw,avg}} = D_{\max} \cdot I_{L,\max} = 0.8 \times 5 = 4 \text{ A}$$

$I_{\text{switch, avg}} = 4 \text{ A}$

Diode average current

Diode conducts for $(1 - D)$ portion:

$$I_{D,\text{avg}} = (1 - D_{\max})I_{L,\max}$$

$$I_{D,\text{avg}} = 0.2 \times 5 = 1 \text{ A}$$

$I_{\text{diode, avg}} = 1 \text{ A}$

—

3. Minimum Inductance

Inductor ripple for a boost converter:

$$\Delta I_L = \frac{V_{\text{in}} D}{L f_s}$$

Worst case ripple occurs at:

- minimum $V_{\text{in}} = 100 \text{ V}$ - maximum $D = 0.8$

Solve for L :

$$L = \frac{V_{\text{in}} D}{\Delta I_L f_s}$$

$$L = \frac{100 \times 0.8}{0.2 \times 50 \times 10^3}$$

$$L = \frac{80}{10,000} = 8 \text{ mH}$$

$L_{\min} = 8 \text{ mH}$

—

4. Minimum Output Capacitance

Voltage ripple expression:

$$\Delta V_o = \frac{I_o D}{C f_s}$$

Worst case ripple occurs at $D_{\max} = 0.8$:

$$C = \frac{I_o D_{\max}}{\Delta V_o f_s}$$

$$C = \frac{1 \times 0.8}{10 \times 50 \times 10^3}$$

$$C = \frac{0.8}{500,000} = 1.6 \mu\text{F}$$

$$\boxed{C_{\min} = 1.6 \mu\text{F}}$$

Final Answers

$$\boxed{I_{\text{switch, avg}} = 4 \text{ A}}$$

$$\boxed{I_{\text{diode, avg}} = 1 \text{ A}}$$

$$\boxed{L_{\min} = 8 \text{ mH}}$$

$$\boxed{C_{\min} = 1.6 \mu\text{F}}$$

$$\boxed{D \in [0.5, 0.8]}$$

Q. Define and explain CCM and compare it with DCM.

Answer:

CCM (Continuous Conduction Mode):

Continuous Conduction Mode is a mode of operation in a DC–DC converter in which the inductor current never falls to zero during the entire switching cycle. In CCM, energy transfer from input to output is continuous, resulting in lower current ripple and better efficiency under medium to high load conditions.

$$i_L(t) > 0 \quad \forall t$$

Comparison between CCM and DCM:

Parameter	CCM	DCM
<i>Inductor Current</i>	<i>Never becomes zero</i>	<i>Falls to zero during part of the cycle</i>
<i>Ripple Current</i>	<i>Low</i>	<i>High</i>
<i>Efficiency</i>	<i>Higher at medium – high load</i>	<i>Better at light load</i>
<i>Output Ripple</i>	<i>Low</i>	<i>Relatively high</i>
<i>Control</i>	<i>More complex</i>	<i>Easier but less stable at high load</i>

Q. Prove that in a buck–boost converter the minimum required output capacitor is

$$C_{\min} = \frac{I_0 d_{\min}}{f_s \Delta V_0}.$$

Answer:

In a buck–boost DC–DC converter, during the switch ON interval (duty ratio D), the output is not connected to the inductor. Therefore, the output capacitor alone supplies the load current I_0 . Let switching frequency be f_s and the allowable peak–to–peak output ripple be ΔV_0 .

$$T_s = \frac{1}{f_s}$$

The capacitor discharge time equals the ON time of the switch:

$$t_{\text{supply}} = D \cdot T_s = \frac{D}{f_s}$$

The charge lost by the capacitor when it supplies the load is:

$$\Delta Q = I_0 \cdot t_{\text{supply}} = I_0 \cdot \frac{D}{f_s}$$

The voltage ripple is related to capacitor value by:

$$\Delta V_0 = \frac{\Delta Q}{C}$$

Substituting ΔQ :

$$\Delta V_0 = \frac{I_0 \cdot \left(\frac{D}{f_s}\right)}{C}$$

Rearranging for C :

$$C = \frac{I_0 D}{f_s \Delta V_0}$$

To ensure safe operation over the minimum duty range, we set $D = d_{\min}$:

$$C_{\min} = \frac{I_0 d_{\min}}{f_s \Delta V_0}$$

Thus, the minimum capacitance required in a buck–boost converter depends on load current, duty ratio, switching frequency, and allowable voltage ripple.

Q. Prove the expression for the output voltage of a DC–DC buck–boost converter in terms of input voltage V_{in} and duty cycle D .

Answer:

For a buck–boost converter operating in continuous conduction mode (CCM), the inductor current ripple over one switching period must satisfy volt–second balance. Let the switch remain ON for a duration DT_s and OFF for a duration $(1 - D)T_s$, where D is the duty cycle.

$$T_s = \frac{1}{f_s}$$

During the ON interval, the inductor is connected to the input source:

$$V_L = V_{in} \quad \text{for } DT_s$$

During the OFF interval, the inductor releases energy to the output through the diode:

$$V_L = -(V_0) \quad \text{for } (1 - D)T_s$$

For steady–state operation, the average voltage across the inductor over one complete switching cycle must be zero (volt–second balance principle):

$$V_{in}(DT_s) + (-V_0)((1 - D)T_s) = 0$$

Divide by T_s :

$$V_{in}D - V_0(1 - D) = 0$$

Expand and rearrange:

$$V_{in}D = V_0 - V_0D$$

$$V_{in}D + V_0D = V_0$$

$$D(V_{in} + V_0) = V_0$$

$$\boxed{V_0 = \frac{D}{1 - D} V_{in}}$$

Thus the output voltage of a buck–boost DC–DC converter is directly proportional to duty cycle D and input voltage V_{in} , and increases non-linearly as D approaches 1.

Q. Prove that in a buck–boost converter the minimum duty cycle is given by

$$d_{\min} = \frac{V_0}{V_{in_{\max}} + V_0}.$$

Answer:

For an ideal buck–boost converter operating in continuous conduction mode (CCM), the relationship between output voltage and input voltage is

$$V_0 = -\frac{D}{1-D}V_{in}.$$

Taking magnitude,

$$|V_0| = \frac{D}{1-D}V_{in}.$$

Rearranging,

$$|V_0|(1-D) = DV_{in}$$

$$|V_0| - |V_0|D = DV_{in}$$

$$|V_0| = D(V_{in} + |V_0|).$$

Solving for D ,

$$D = \frac{|V_0|}{V_{in} + |V_0|}.$$

Since output voltage magnitude is normally written simply as V_0 ,

$$D = \frac{V_0}{V_{in} + V_0}.$$

The minimum duty cycle occurs when the input voltage is maximum, therefore

$$d_{\min} = D \Big|_{V_{in}=V_{in\max}}$$

$$\boxed{d_{\min} = \frac{V_0}{V_{in\max} + V_0}}.$$

Thus, the minimum operating duty of a buck–boost converter depends on the ratio between output voltage and the maximum input voltage.

3 Transformer

3.1 emf equation of transformer

Consider a transformer with N_1 turns on the primary winding. Let the mutual flux linking the core be sinusoidal and given by

$$\phi(t) = \phi_{\max} \sin \omega t$$

where

$$\omega = 2\pi f$$

Step 1: Induced EMF in the winding

According to Faraday's law of electromagnetic induction, the instantaneous induced emf in the primary winding is

$$e(t) = -N_1 \frac{d\phi(t)}{dt}$$

Substituting $\phi(t)$,

$$e(t) = -N_1 \frac{d}{dt} (\phi_{\max} \sin \omega t)$$

$$e(t) = -N_1 \phi_{\max} \omega \cos \omega t$$

Step 2: Maximum value of induced EMF

The maximum value of induced emf occurs when $\cos \omega t = 1$:

$$E_{\max} = N_1 \phi_{\max} \omega$$

Substituting $\omega = 2\pi f$,

$$E_{\max} = 2\pi f N_1 \phi_{\max}$$

Step 3: RMS value of induced EMF

The rms value of a sinusoidal emf is related to its maximum value as

$$E_{\text{rms}} = \frac{E_{\max}}{\sqrt{2}}$$

Therefore,

$$E_{\text{rms}} = \frac{2\pi f N_1 \phi_{\max}}{\sqrt{2}}$$

Rearranging,

$$\boxed{\phi_{\max} = \frac{E_{\text{rms}}}{\sqrt{2} \pi f N_1}}$$

Step 4: For an ideal transformer

For an ideal transformer, the rms induced emf in the primary winding is approximately equal to the applied rms primary voltage, i.e.,

$$E_{\text{rms}} \approx V_1$$

Hence,

$$\phi_{\text{max}} = \frac{V_1}{\sqrt{2} \pi f N_1}$$

Therefore, the emf per turn is

$$\frac{E}{N} = 4.44 f \phi_{\text{max}}$$

Alternatively, starting from the instantaneous flux:

$$\phi(t) = \phi_{\text{max}} \sin \omega t$$

$$e(t) = -N \frac{d\phi}{dt}$$

$$E_{\text{max}} = 2\pi f N \phi_{\text{max}}$$

$$E_{\text{rms}} = \frac{E_{\text{max}}}{\sqrt{2}}$$

$$\Rightarrow \frac{E_{\text{rms}}}{N} = \frac{2\pi f \phi_{\text{max}}}{\sqrt{2}} = 4.44 f \phi_{\text{max}}$$

Hence, emf per turn is

$$\frac{E}{N} = 4.44 f \phi_{\text{max}}$$

Physical Meaning of EMF per Turn

The *emf per turn* of a transformer is the voltage induced in a single turn of the winding due to the time-varying magnetic flux in the core. It indicates how effectively the alternating core flux is converted into electrical voltage.

Mathematically, the emf per turn is given by

$$\frac{E}{N} = 4.44 f \phi_{\text{max}}$$

where f is the supply frequency and ϕ_{max} is the maximum value of magnetic flux in the core. Thus, the emf per turn depends only on the frequency and the core flux, and is independent of the number of turns.

Physical interpretation:

- **Measure of core flux level:** Since the emf per turn is directly proportional to ϕ_{max} , it serves as an indicator of the magnetic loading of the transformer core. A higher emf per turn implies a higher flux density, which may drive the core towards magnetic saturation.

- **Design constraint:** For a given core material and operating frequency, there exists a maximum permissible emf per turn. Exceeding this limit results in increased core losses, excessive magnetizing current, and possible saturation of the core.
- **Uniformity across windings:** In an ideal transformer, all windings link the same core flux. Hence, the emf per turn is the same for all windings:

$$\frac{E_1}{N_1} = \frac{E_2}{N_2} = \text{constant}$$

- **Role in transformer design:** In practical design, the emf per turn is selected first based on core material and frequency. The required number of turns for a given voltage rating is then obtained as

$$N = \frac{E}{\text{emf per turn}}$$

- **Operational significance:** Maintaining an appropriate emf per turn ensures efficient operation, prevents core saturation, and limits iron losses, thereby improving the overall performance and reliability of the transformer.

Hence, the emf per turn represents the magnetic stress on the transformer core and is a key parameter in transformer design and analysis.

Distribution transformer

A distribution transformer is designed to deliver electrical energy to end users under continuously varying load conditions. The design and operation of such transformers are governed by practical application of transformer principles.

1. Operation at Nearly Constant Voltage

Distribution transformers are connected close to the consumer end and must maintain the rated supply voltage within permissible limits despite load variations. The induced emf in the secondary winding is given by

$$E = 4.44fN\phi_{\max}$$

For a constant supply frequency f and controlled maximum flux $\phi_{\max}$, the output voltage remains nearly constant. Therefore, the number of turns is selected such that voltage regulation is minimal under typical load conditions.

2. Rating in kVA Rather Than kW

The losses in a transformer depend on voltage and current, not directly on the load power factor.

- **Core (iron) losses** depend on voltage and frequency and remain nearly constant.
- **Copper losses** depend on current and vary as I^2R .

Since power factor is decided by the consumer load and not by the transformer itself, distribution transformers are rated in **kVA** rather than kW.

3. Influence of Flux Density and Turns on Light-Load Efficiency

Distribution transformers operate for long durations under light-load conditions. The efficiency under such conditions is mainly influenced by core losses.

- Lower flux density reduces hysteresis and eddy current losses.
- A higher number of turns reduces flux density for a given voltage.

Since core losses dominate at light loads, the transformer is designed with low-loss core material and appropriate turns per volt to achieve high all-day efficiency.

By applying transformer emf relations and loss characteristics, it is evident that distribution transformers are optimized for constant voltage operation, kVA rating, and high efficiency over long operating hours with varying loads.

Dot Convention in Magnetic Circuits

The *dot convention* is a symbolic method used in magnetic circuits and coupled inductor or transformer representations to indicate the *relative polarity* of induced voltages and the *direction of mutual flux linkage* between windings.

Basic principle:

When two or more coils are magnetically coupled, the dot marks are placed on the terminals of the windings such that:

- If current enters the dotted terminal of one coil, the induced voltage in the other coil is positive at its dotted terminal.
- If current leaves the dotted terminal of one coil, the induced voltage in the other coil is negative at its dotted terminal.

Thus, the dot convention indicates whether the mutually induced emf in one coil *aids* or *opposes* the self-induced emf of the other coil.

Mathematical interpretation:

For two magnetically coupled coils with self-inductances L_1 and L_2 , and mutual inductance M , the voltage equations using dot convention are written as

$$v_1 = L_1 \frac{di_1}{dt} \pm M \frac{di_2}{dt}$$

$$v_2 = L_2 \frac{di_2}{dt} \pm M \frac{di_1}{dt}$$

The sign of the mutual term depends on the direction of currents with respect to the dotted terminals:

- The mutual term is taken as **positive** if both currents enter (or leave) the dotted terminals simultaneously.
- The mutual term is taken as **negative** if one current enters the dotted terminal while the other leaves its dotted terminal.

Physical significance:

The dot convention represents the orientation of windings on a common magnetic core. It identifies the direction of the mutual magnetic flux produced by one winding and linking the other. Proper application of dot convention ensures correct prediction of induced voltages, power flow, and phase relationships in transformers and coupled inductors.

Importance in circuit analysis:

- It helps in correctly writing mesh or nodal equations for magnetically coupled circuits.
- It avoids sign ambiguity in mutual inductance terms.
- It ensures correct polarity of induced emf in transformer and coupled inductor analysis.

Hence, the dot convention is an essential tool for determining voltage polarity and mutual interaction in magnetic circuits and transformer analysis.

Example: Application of Dot Convention

Consider two magnetically coupled coils with self-inductances L_1 and L_2 , and mutual inductance M . The coils are wound on a common magnetic core and their dotted terminals are as shown.

Let the current i_1 enter the dotted terminal of coil 1, and the current i_2 enter the dotted terminal of coil 2.

Voltage equations using dot convention:

According to Faraday's law, the voltage across each coil consists of a self-induced term and a mutually induced term.

For coil 1:

$$v_1 = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

For coil 2:

$$v_2 = L_2 \frac{di_2}{dt} + M \frac{di_1}{dt}$$

The mutual inductance terms are *positive* because both currents enter the dotted terminals simultaneously. Hence, the induced emfs assist each other.

Case 2: Reversal of current direction

If i_1 enters the dotted terminal of coil 1, but i_2 leaves the dotted terminal of coil 2, then the polarity of the mutually induced emf in coil 2 reverses.

The voltage equations become:

$$v_1 = L_1 \frac{di_1}{dt} - M \frac{di_2}{dt}$$

$$v_2 = L_2 \frac{di_2}{dt} - M \frac{di_1}{dt}$$

Here, the mutual inductance terms are *negative* because one current enters a dotted terminal while the other leaves its dotted terminal.

Physical interpretation:

- When currents enter (or leave) the dotted terminals together, the mutual fluxes aid each other and the mutual emf is additive.
- When one current enters a dotted terminal and the other leaves its dotted terminal, the mutual fluxes oppose each other and the mutual emf is subtractive.

Thus, the dot convention directly determines the sign of the mutual inductance terms in the voltage equations of magnetically coupled circuits.

Numerical Example Based on Dot Convention

Two magnetically coupled coils have self-inductances

$$L_1 = 2 \text{ H}, \quad L_2 = 8 \text{ H}$$

and mutual inductance

$$M = 1 \text{ H}.$$

The currents in the coils are

$$i_1(t) = 5t \text{ A}, \quad i_2(t) = 2t \text{ A}.$$

Both currents enter the dotted terminals of their respective coils.

Determine the instantaneous voltages $v_1(t)$ and $v_2(t)$.

Solution:

The voltage equations for magnetically coupled coils using dot convention are

$$\begin{aligned} v_1 &= L_1 \frac{di_1}{dt} + M \frac{di_2}{dt} \\ v_2 &= L_2 \frac{di_2}{dt} + M \frac{di_1}{dt} \end{aligned}$$

since both currents enter the dotted terminals.

Step 1: Compute current derivatives

$$\begin{aligned} \frac{di_1}{dt} &= 5 \text{ A/s} \\ \frac{di_2}{dt} &= 2 \text{ A/s} \end{aligned}$$

Step 2: Substitute values

$$\begin{aligned} v_1 &= (2)(5) + (1)(2) \\ v_1 &= 10 + 2 = 12 \text{ V} \end{aligned}$$

$$\begin{aligned} v_2 &= (8)(2) + (1)(5) \\ v_2 &= 16 + 5 = 21 \text{ V} \end{aligned}$$

Final Answer:

$$\boxed{v_1(t) = 12 \text{ V}}, \quad \boxed{v_2(t) = 21 \text{ V}}$$

Note:

If one of the currents had left its dotted terminal instead of entering, the mutual inductance terms would be negative, and the voltages would change accordingly.

Experimental Determination of Dot Polarity of a Transformer

When the dot markings of a transformer windings are unknown, the relative polarity can be determined experimentally using a low-voltage test. Two standard methods are commonly used.

Method 1: DC Pulse (Kick) Test

Procedure:

1. Identify the primary and secondary windings of the transformer using a continuity test or resistance measurement.
2. Connect one winding (taken as the primary) to a low-voltage DC source (typically 1.5–6 V) through a switch.
3. Connect a sensitive center-zero analog voltmeter or galvanometer across the other winding (secondary).
4. Momentarily close the switch to apply the DC voltage to the primary winding. Observe the direction of deflection in the galvanometer.
5. If the galvanometer shows a positive deflection at the instant of switching, the terminal of the secondary connected to the positive terminal of the meter has the same polarity as the primary terminal connected to the positive terminal of the DC source.
6. Mark these terminals as the *dotted ends* of the respective windings.
7. Repeat the test to confirm the observed polarity.

Method 2: AC Voltage (Additive–Subtractive) Test

Procedure:

1. Apply a low AC voltage (typically 6–12 V) to the primary winding of the transformer.
2. Measure the open-circuit voltage across the secondary winding using an AC voltmeter.
3. Connect one terminal of the primary winding to one terminal of the secondary winding to form a series connection.
4. Measure the voltage across the free terminals of the series-connected windings.
5. If the measured voltage is approximately equal to the *sum* of the individual voltages, the connected terminals are of opposite polarity.
6. If the measured voltage is approximately equal to the *difference* of the individual voltages, the connected terminals are of the same polarity.

7. Mark the terminals which produce subtractive voltage when connected as the *dotted terminals*.

Precautions:

- Use only low voltage during the experiment.
- Keep the transformer unloaded while performing the test.
- Avoid applying rated voltage until the polarity is confirmed.

Battery Specifications

Battery specifications define the electrical characteristics and performance limits of a battery. The important battery specifications are as follows:

1. **Nominal Voltage (V):** The rated voltage of the battery under normal operating conditions.
2. **Capacity (Ampere-hour, Ah):** Battery capacity indicates the amount of charge a battery can deliver. It is defined as

$$\text{Capacity (Ah)} = I \times t$$

where I is the discharge current in amperes and t is the discharge time in hours.

For example, a 100 Ah battery can deliver:

$$5 \text{ A for 20 h} \quad \text{or} \quad 10 \text{ A for 10 h.}$$

3. **Energy Capacity (Watt-hour, Wh):** The total energy stored in the battery, given by

$$\text{Energy (Wh)} = V \times \text{Ah}$$

4. **Rated Current:** The maximum continuous current that the battery can safely supply.
5. **Internal Resistance:** The effective resistance inside the battery that causes voltage drop and power loss.
6. **Charge and Discharge Rate (C-rate):** Indicates how fast the battery is charged or discharged relative to its capacity.

$$1C = \text{current equal to the Ah rating}$$

7. **Cycle Life:** The number of complete charge–discharge cycles a battery can undergo before its capacity reduces to a specified value (typically 80%).
8. **Efficiency:** Ratio of output energy to input energy during charging and discharging.

Among all specifications, Ampere-hour (Ah) rating is the most commonly used parameter to indicate the charge storage capability of a battery.

Difference Between Ampere-hour (Ah) and Watt-hour (Wh) for Lithium Batteries

In lithium batteries, both Ampere-hour (Ah) and Watt-hour (Wh) ratings are used to specify battery capacity, but they represent different physical quantities.

1. Ampere-hour (Ah):

The Ampere-hour rating indicates the *charge capacity* of a battery. It specifies how much electric charge the battery can deliver over time.

$$\text{Ah} = I \times t$$

where I is the discharge current (A) and t is the discharge time (h).

Limitation: Ah rating alone does not account for the battery voltage, hence it does not directly represent the stored energy.

2. Watt-hour (Wh):

The Watt-hour rating indicates the *energy capacity* of a battery. It represents the total usable energy stored in the battery.

$$\text{Wh} = V \times \text{Ah}$$

where V is the nominal battery voltage.

3. Importance in Lithium Batteries:

Lithium batteries are available at different nominal voltages (e.g., 3.6 V, 7.2 V, 48 V). Therefore, two batteries with the same Ah rating can store different amounts of energy.

Example:

Battery A: 10 Ah, 12 V $\Rightarrow$ 120 Wh

Battery B: 10 Ah, 48 V $\Rightarrow$ 480 Wh

Although both batteries have the same Ah rating, Battery B stores four times more energy.

4. Practical Interpretation:

- Ah rating is useful for estimating *current delivery duration*.
- Wh rating is essential for comparing *actual energy storage* and runtime.

For lithium batteries, Wh is a more accurate and universal specification than Ah, especially when comparing batteries with different voltage ratings.

Why Airlines Specify Watt-hour (Wh) Instead of Ampere-hour (Ah)

Airlines specify battery limits in terms of **Watt-hour (Wh)** rather than **Ampere-hour (Ah)** because Wh directly represents the *total energy content* of a battery, which is the key factor related to safety risks such as overheating and fire hazards.

1. Energy Determines Safety Risk

The potential severity of a battery fire depends on the total energy stored in the battery.

Watt-hour rating measures this energy:

$$\text{Wh} = V \times \text{Ah}$$

A higher Wh value means more stored energy and therefore a higher fire risk.

2. Ah Does Not Account for Voltage

Ampere-hour rating indicates only the charge capacity. Two lithium batteries can have the same Ah rating but very different voltages and energy contents.

Example:

Battery A: 5 Ah, 3.7 V $\Rightarrow$ 18.5 Wh

Battery B: 5 Ah, 22.2 V $\Rightarrow$ 111 Wh

Both batteries have identical Ah ratings, but Battery B stores significantly more energy.

3. Uniform Standard Across Battery Types

Lithium batteries are manufactured with various voltage levels by connecting cells in series. Wh provides a common basis for comparison regardless of:

- Number of cells
- Cell chemistry
- Nominal voltage

4. Regulatory Limits Are Energy-Based

Aviation authorities (IATA, ICAO, FAA) specify limits such as:

- Batteries ≤ 100 Wh: allowed in carry-on baggage
- Batteries 100–160 Wh: allowed with airline approval

These limits are based on total energy, not charge capacity.

Airlines specify **Wh instead of Ah** because Wh:

- Accurately represents total stored energy
- Directly correlates with fire and thermal runaway risk
- Allows fair comparison across different battery voltages

Hence, Wh is the safest and most meaningful metric for aviation safety regulations.

Problem 1:

A lithium-ion battery is rated at 12 Ah and 18.5 V.

Solution:

$$\text{Energy} = V \times \text{Ah} = 18.5 \times 12 = 222 \text{ Wh}$$

Since $222 \text{ Wh} > 100 \text{ Wh}$, this battery is **not allowed** in carry-on baggage without airline approval.

Problem 2:

Two batteries are rated as follows:

Battery A: 8 Ah, 3.7 V

Battery B: 4 Ah, 7.4 V

Solution:

$$\text{Energy}_A = 8 \times 3.7 = 29.6 \text{ Wh}$$

$$\text{Energy}_B = 4 \times 7.4 = 29.6 \text{ Wh}$$

Both batteries store **equal energy**.

Problem 3:

A power bank is rated at 20 Ah and 5 V.

Solution:

$$\text{Energy} = 5 \times 20 = 100 \text{ Wh}$$

This power bank is **allowed** in airline carry-on baggage without approval.

Problem 4:

A drone battery has an energy rating of 150 Wh and a nominal voltage of 22.2 V.

Solution:

$$\text{Ah} = \frac{\text{Wh}}{V} = \frac{150}{22.2} = 6.76 \text{ Ah}$$

Problem 5:

A laptop battery consists of 6 cells in series, each rated at 3.7 V and 3 Ah.

Solution:

Total voltage:

$$V_{\text{total}} = 6 \times 3.7 = 22.2 \text{ V}$$

Total energy:

$$\text{Wh} = 22.2 \times 3 = 66.6 \text{ Wh}$$

Problem 6:

An EV battery pack is rated at 48 V and 100 Ah.

Solution:

Total energy:

$$\text{Energy} = 48 \times 100 = 4800 \text{ Wh} = 4.8 \text{ kWh}$$

Driving range:

$$\text{Range} = \frac{4800}{200} = 24 \text{ km}$$

Problem 7:

An airline permits batteries up to 160 Wh with approval. A battery is rated at 14.8 V and 10 Ah.

Solution:

$$\text{Energy} = 14.8 \times 10 = 148 \text{ Wh}$$

Since $148 \text{ Wh} < 160 \text{ Wh}$, airline approval is **required**.

Problem 8:

A UPS battery is rated at 24 V and 50 Ah supplying a 600 W load.

Solution:

Energy stored:

$$\text{Wh} = 24 \times 50 = 1200 \text{ Wh}$$

Backup time:

$$t = \frac{1200}{600} = 2 \text{ h} = 120 \text{ minutes}$$

Key Exam Formula:

$\text{Wh} = V \times \text{Ah}$

Why Ampere-hour (Ah) Is Still Specified Even Though Watt-hour (Wh) Is More Important

Although **Watt-hour (Wh)** is the correct measure of total energy stored in a battery, **Ampere-hour (Ah)** is still specified because it provides important and complementary information related to battery operation and application requirements.

1. Ah Indicates Charge Delivery Capability

Ah rating specifies the total electric charge a battery can supply:

$$\text{Ah} = I \times t$$

It helps designers estimate how long a battery can supply a given current.

2. Ah Is Essential for Current-Based Design

Many electrical systems are designed based on current requirements rather than power. Examples include:

- Motor drive current limits
- Inverter and converter current ratings
- Fuse and cable sizing

Ah rating directly relates to how much current can be delivered over time.

3. Ah Helps in Cell Matching and Pack Design

In battery packs, cells connected in series must have identical Ah ratings. This ensures:

- Uniform charging and discharging
- Prevention of cell imbalance
- Improved cycle life

4. Voltage Is Often Fixed in a Given Application

In many applications (e.g., mobile phones, laptops), the battery voltage is standardized. In such cases:

$$\text{Wh} \propto \text{Ah}$$

Thus, Ah becomes a convenient indicator of runtime.

5. Battery Management System (BMS) Uses Ah

State-of-charge (SOC) estimation is commonly based on coulomb counting:

$$\text{SOC} \propto \int I \, dt$$

which directly relates to Ah, not Wh.

6. Practical Example

Two batteries with identical Wh but different voltages will have different Ah ratings. This affects:

- Maximum current capability
- Internal resistance losses

- Thermal performance
- **Wh** specifies *how much energy* a battery stores.
- **Ah** specifies *how much charge* and *current capability* the battery has.

Both Ah and Wh are necessary specifications, serving **different but complementary purposes** in battery design, application, and safety assessment.

State of Charge (SOC) and State of Discharge (SOD)

1. State of Charge (SOC):

State of Charge (SOC) represents the amount of charge remaining in a battery relative to its rated capacity. It is usually expressed as a percentage.

$$\text{SOC}(\%) = \frac{\text{Remaining charge (Ah)}}{\text{Rated capacity (Ah)}} \times 100$$

Interpretation:

- $\text{SOC} = 100\% \rightarrow$ battery fully charged
- $\text{SOC} = 0\% \rightarrow$ battery fully discharged

2. State of Discharge (SOD):

State of Discharge (SOD) indicates how much of the battery capacity has already been used.

$$\text{SOD}(\%) = \frac{\text{Discharged capacity (Ah)}}{\text{Rated capacity (Ah)}} \times 100$$

Relation between SOC and SOD:

$$\boxed{\text{SOC} + \text{SOD} = 100\%}$$

3. Importance of SOC and SOD:

1. **Battery Protection:** Prevents overcharging ($\text{SOC} > 100\%$) and deep discharge ($\text{SOC} \approx 0\%$), which degrade battery life.
2. **Energy Management:** Used in EVs, solar systems, and UPS to schedule charging and load sharing.
3. **Range and Runtime Estimation:** SOC is used to estimate remaining operating time or vehicle range.
4. **Battery Health and Safety:** Maintaining SOC within safe limits improves cycle life and prevents thermal runaway.

4. Relation of SOC and SOD with Ampere-hour (Ah):

SOC and SOD are directly related to Ah through *coulomb counting*.

$$\text{Remaining Ah} = \text{Rated Ah} - \int I(t) dt$$

Thus,

$$\text{SOC} = 100 \left(1 - \frac{\int I(t) dt}{\text{Rated Ah}} \right)$$

Example:

A battery rated at 50 Ah has delivered 15 Ah to the load.

$$\text{SOC} = \frac{50 - 15}{50} \times 100 = 70\%$$

$$\text{SOD} = \frac{15}{50} \times 100 = 30\%$$

5. Practical Implementation in BMS:

Battery Management Systems (BMS) estimate SOC using:

- Coulomb counting (Ah-based)
- Open-circuit voltage (OCV) method
- Kalman filtering and hybrid techniques

SOC and SOD are critical indicators of battery usage and health. They are directly related to Ah, making Ah an essential parameter for monitoring, control, and protection of batteries in practical systems.

Problem 1 A lithium-ion battery is rated at 60 Ah. The battery has already delivered 18 Ah to the load.

Solution:

$$\text{SOC} = \frac{60 - 18}{60} \times 100 = 70\%$$

Problem 2: SOC After Constant Current Discharge

A 100 Ah battery is discharged at a constant current of 5 A for 6 hours.

Solution:

Discharged capacity:

$$Q = I \times t = 5 \times 6 = 30 \text{ Ah}$$

SOC:

$$\text{SOC} = \frac{100 - 30}{100} \times 100 = 70\%$$

Problem 3: SOC During Variable Current Discharge

A 40 Ah battery supplies a current given by

$$I(t) = 2 + 0.5t \quad (\text{A})$$

where t is in hours, for a duration of 4 hours.

Solution:

Discharged charge:

$$Q = \int_0^4 (2 + 0.5t) dt$$

$$Q = [2t + 0.25t^2]_0^4 = 8 + 4 = 12 \text{ Ah}$$

SOC:

$$\text{SOC} = \frac{40 - 12}{40} \times 100 = 70\%$$

Problem 4: SOC During Charging

A 50 Ah battery initially has an SOC of 40%. It is charged at a constant current of 5 A for 4 hours with a charging efficiency of 90%.

Solution:

Initial charge:

$$Q_{\text{initial}} = 0.4 \times 50 = 20 \text{ Ah}$$

Effective charged capacity:

$$Q_{\text{charge}} = 5 \times 4 \times 0.9 = 18 \text{ Ah}$$

Final charge:

$$Q_{\text{final}} = 20 + 18 = 38 \text{ Ah}$$

Final SOC:

$$\text{SOC} = \frac{38}{50} \times 100 = 76\%$$

Problem 5: SOC in an Electric Vehicle

An EV battery pack is rated at 60 kWh. The vehicle has consumed 15 kWh during a trip.

Solution:

Remaining energy:

$$E_{\text{remaining}} = 60 - 15 = 45 \text{ kWh}$$

SOC:

$$\text{SOC} = \frac{45}{60} \times 100 = 75\%$$

Problem 6: SOC Estimation Using Coulomb Counting

A 80 Ah battery supplies a current of 10 A for 2 hours and then 5 A for 4 hours.

Solution:

Total discharged capacity:

$$Q = (10 \times 2) + (5 \times 4) = 20 + 20 = 40 \text{ Ah}$$

SOC:

$$\text{SOC} = \frac{80 - 40}{80} \times 100 = 50\%$$

SOC estimation is generally based on:

$$\boxed{\text{SOC} = 100 \left(1 - \frac{\int I(t) dt}{\text{Rated Ah}} \right)}$$

and is commonly implemented using coulomb counting in Battery Management Systems (BMS).

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